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"INTERACTIVE HEALTH BOT"

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in
ELECTRONICS & COMMUNICATION ENGINEERING
by

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B.N.M. Institute of Technology

Approved by AICTE, Affiliated to VTU, Accredited as Grade A Institution by NAAC.
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CERTIFICATE

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ABSTRACT

We all encounter small accidents and injuries in our daily lives. Often, people fall sick with the change in season, particularly elderly people. Is it feasible and necessary to run to a doctor every time for a simple flu or an injury? We would say 'NO'. But some measures have to be taken to treat these basic injuries and flu and identify its severity. To deal with this problem, we have come up with an innovative solution of a bot which will help people to get all the first aid help they need and also get information about the probability of the diseases they might have. In this era, health monitoring devices are the need of the hour as there is an increase in diseases of different kinds. For certain diseases people have to visit the doctor as soon as possible and take treatment to stop it or slowdown from making more severe damage. Conventionally when people fall sick, they try to be the doctor themselves and ignore their issue by swallowing few tablets to get rid of the ailment for the time being. But these issues might pile up causing them bigger trouble. To help people to get timely and proper diagnosis and help them to take more care about their health, we propose the idea of a healthaiddevice.

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CHAPTER 1

INTRODUCTION

Chapter – 1

INTRODUCTION

1.1 Introduction

The access to medical care is unacceptably low worldwide in spite of the increasing demand for medical care due to population aging and increasing burden of diseases. Especially in resource limited places, patients do not have access to medical care in a timely manner. Thus, tools to help patients with self-diagnosis and self-triage is urgently needed to mitigate the shortage of medicalcareresources.

Health monitoring devices are the need of the hour in this era of increasing diseases of different kinds. All the diseases have to be treated as soon as possible and a remedy has to be provided to stop it from making moreseveredamage.

Conventionally when people fall sick, they try to be the doctor themselves and ignore their issue by swallowing few tablets to get rid of it for the time being. But these issues might pile up later causing them bigger trouble. To help people get sooner diagnostics and help them take more care about their health, we have come up with this idea of health-aid device called as the –Interactivehealthbot'.

People use the internet to research about their health concerns. Thousands of people regularly use the internet to self-diagnose their ailments, both for non-urgent symptoms and for urgent symptoms such as chest pain. While there is a wealth of online resources to learn about specific conditions, self-diagnosis usually starts with search engines like Google. However, internet search engines can lead users to confusing and sometimes unsubstantiated information, and people with urgent symptoms may not be directed to seek emergent care. Hence, our aim is to attempt to more effectively provide a potential diagnosis for patients and direct them to theappropriate caresetting.

The device serves to facilitate self-diagnosis and to assist with triage. The diagnosis deals with typically educating patients on the range of diagnoses that might fit their symptoms. The triage informs patients whether they should seek care at all and, if so, where and with what urgency (that is, emergently or within afewdays).

For patients with a non-emergent problem that does not require a medical visit, our device can reassure and recommend they stay home and prevent them from using unnecessary antibiotics.

Reducing the number of visits saves patient's time and money, deters overprescribing of antibiotics. Logging of all the diagnosis and the symptoms that led to it and other relevant information would help in statistical analyzing and interpretation.

This device focuses on providing health related aid to the society. It is a hardware device which is interfaced with sensors. The device takes voice commands and processes those commands to provide details about a disease when the symptoms are provided to the device. It goes by asking the users a series of questions about their symptoms or requires users to input details about their symptoms themselves. The symptoms are classified into a disease using a decision tree and an audio and text output is provided to the end user at each and every step until it drills down to the closest disease that is matched. The output includes the disease related information, and any of the common treatment procedures if it is not a serious one. The temperature sensor acts as one of the symptoms measuring interfaces that are present on the device.

Basically, it makes a diagnosis of a disease based on the data on the prevalence of disease and its sensitivity and specificity of symptoms. In the future, such devices would help keeping up the quality of primary care.

1.2 Motivation

Healthcare is one of the highest, ever-green, growing sectors in the world. There are many opportunities for players in the medical device industry. The intention of universal healthcare is to make it accessible to all individuals and families in a community. Our interest in this project lies in addressing the health and well-being of people. Our aim is to educate or inform the patient by extracting their symptoms and informing them the possible anomaly and the level of attention that might have to begiven to it.

Also, it is increasingly common for patients to query their symptoms through a search engine before approaching any medical professional. Hence building a symptom-diagnosis device is a significant research problem for personalized care.

Although many research and study has been done in this area our goal in this project is to integrate many features in a single handy device. A device which could have the easiest user interface with least user inputs and dependency, giving out a close to accurate prognosis prediction and enlightening the user withthe same.

Further,collecting the diagnosis and related data, from our device and maintaining the user's privacy at the same time would be a good pitch in for analyzing and statistics. However, we would like to give a note of caution, since the tools could go wrong.

All in all, we sought to create a device that should be the first option for any medical condition that is not an emergency.

1.3 Problem Statement

We all know how important first aid is in providing medical assistance to a person suffering from a medical condition. Similarly, first advice and first understanding of the symptoms is equally important.

Keeping our target market in mind, it is only natural that we include a speech-to- text feature into our project. The speech-to-text conversion will be done using natural language processing. The first step of speech recognition is, of course, speech. Speech from physical sound must be converted into intermediate formats such as electrical signal to ultimately get our audio input as a digital data. Once digitized, several models can be used to transcribe theaudio to text.

After the speech-to-text conversion, keywords are extracted from the text obtained. Identification of the keyword helps the device to move forward in the project. Thus, identified keyword is mapped with the database and using classification tree the disease is recognized.

Additionally, sensors are interfaced which also provide vital information about the patient. The whole system is coded onto a raspberry pi which acts like a standalone device performing all the necessary computations.

Our project proposes to implement a solution to help the user with primary care medical advice connecting symptoms with ailments with complete storing of the diagnosed information in a cloud-based database for any further analytics.

1.4 Objective

The main aim of this project is to develop a device for diagnosing and classifying diseases based on the symptoms they exhibit. The symptoms will be taken in an audio

format through an LCD display screen and RE speakers. The audio is then converted into text format on which NLP techniques would be applied such that the symptoms expressed by the patients correspond to a symptom in our existing symptom database. Further along, the classification will take place using a decision tree called CART which will work in real time as the user inputs more symptoms until they match a disease.

This result would be shared to the user through the LCD display and RE speaker. In addition to this, the classification results along with other details such as time, date etc. will be stored in a cloud database called Google Firebase. This data will be used as a record of patient history and to derive information through statistical analysis on the same.

1.5 Summary

The objective is to assist the people by providing authentic and accurate disease classification, advice and care, by avoiding the trouble of having to visit a hospital and also to provide help in identifying the disease at home. It focuses on improving the overall well-being of the society. To develop a device that would be handy and simple that aid citizens to get better technological medical assistance.The proposed device mainly includes an application based on a classification and regression tree model and takes speech input and provides audio output which will be taken care by the natural language processing. It also includes sensors that measure standard body temperature and pulse.

As an additional scope, there are opportunities for this device not only in homes but also in hospitals, if the concerned doctor is busy or away. We hope that our device meets the mentioned requirements and proves to bring a transformative change in the healthcare technology.

CHAPTER 2

LITERATURE SURVEY

Chapter – 2

LITERATURE SURVEY

2.1 Introduction

A significant portion of medical expenditure in the world is spent on managing chronic disease in the hospital. With rapid increase of aging people with chronic diseases, we need to find smarter ways to manage the health needs without increase of financial burden of hospitals and/or nursing facilities. Examining the person's health condition from the everyday life is more effective thanthe clinical setup.

Recently, scientists have drawn significant research attention on introducing Internet of Things (IoT) in the health-care system, which integrates the Internet with remote monitoring smart sensors and medical device. Acquisition of health-related information by unobtrusive sensing and wearable technologies is considered as a cornerstone in health informatics. The two main acquisition technologies of the health attributes are sensing and imaging. But only the sensor technologies are preferred, especially the unobtrusive sensing and wearable modules for continuous health monitoring.

Several papers were published and proposed different methods for assisting a patient. An interactive robot for reminding medication, a personalized diet and exercise guideline recommendation system, personalized medical assistant based preventive health-care model are some of the many proposed solutions.

2.2 Existing Technologies

One of the proposed hybrid-device includes some of the existing individual modules for monitoring blood glucose, blood-pressure, heart rate, body temperature, and ECG. The most commonly measured vital signs include: Electrocardiography (ECG), blood-glucose, heart rate, blood pressure (BP) and surface body temperature. Sensors that can be integrated with garments or wearable accessories are used and also, they can be organized as stick-on electronictattoos.

This device also suggests regular routine medicine to the patient based on physician's prescription and also advise emergency drug/medicine to the patient depending on the health monitoring attributes. Along with monitoring the health information of the patient, the acquired personalized bio-medical attributes' records are sent to the health center either by cellular or wireless communication media. The proposed device is capable of taking necessary initiatives for emergency medication and treatment in case of unpredictable situation and notifying the patient about regular medication routine like a medical assistant according to physician prescription. It also notifies the prescribed physician about current health condition of the patient. The device may also act as an intelligent agent terminal for patient, which will interact with physician and pharmacy on behalf of the patient, where the physician may prescribe medicine to the patient based on health informatics provided by the sensors of the proposed device and then the agent may track the physicians prescription to place orders automatically to the pharmacy. The device is said to be cost efficient by reducing the number of visits to specialists, avoids symptoms exacerbations, emergency room visits andnursing home admissions.

Limitations / Challenges - User acceptability, portability, reduction of motion artifact, power consumption, self-processing capability, and distributed interference in wireless communication networks. The data acquired from the individual sensor will be large. Accuracy improvement of the health monitoring attributes, miniaturization and unobtrusiveness of the wearable device, reliability of data-communication interface, on-node intelligent data processing and power consumption.[1]

Another paper deals about examining the characteristics of health and fitness apps. A number of health-related voice activated assistants (VAA) that are used in a variety of settings for many different functions, such as home monitoring of symptoms or health education. Some studies have suggested using Amazon Alexa to asses deaf speech, provide task support for individuals with cognitive disabilities, and receive voice input from patients to determine –unexpected changes in mood.Such VAA vendor apps are different from the conventional mobile apps, in that the types of interactions are not fully delineated by voice interface because it does not have the same transparency as interacting with a physical user interface.

There is also no menu of features or functionalities such that the user could understand

the full spectrum of the types of commands one could ask or what kind of information could be provided by the app.

A study was done to conduct the descriptive content analysis based on the information provided by the vendors to determine the types of apps in health and fitness categories for commercially available VAAs. As per the review, Amazon and Google were the only companies with commercially available hands-free VAAs. The selected apps of hands-free voice activated assistants are screened for health and fitness apps and the filtered ones are taken for review. After evaluation it was found that 309 apps were categorized as–health and fitness apps byvendors.Their analysis revealed there was only a limited number of health monitoring apps. This may be, in part, due to the restrictions within Amazon and Google policies for publishing health-related apps. Amazon and Google Homedeviceshaveabuttonthatcanbepressedto–mutethedevicefromlistening,but instruction manuals do not have explicit language describing that the devices are always listening and that they are recording and storing audio. Also, device and usability should be improved, and focus should be given on interactive user-based interfaces. Apps are released, modified, updated, and discontinued on a regular basis. As a result, there may be apps that were reviewed in this study that are no longer available, and there may have been modifications in app descriptions since data extraction. Despite these limitations, the paper contributes towards an understanding of the characteristics of health and fitness apps available for commercially available, hands-freeVAA.[2]

Emad S. Othman proposes, Senior Member IEEE, proposes a Raspberry Pi based personalized voice command system. The microcontroller is the heart of the command system as it is involved in every step of processing data to connecting components together. The Microphone of the raspberry pi is used to take the audio input of the sound. This audio input when further passed through the system would be searched for keywords. These keywords are essential for the functioning of the device as the modules work on the essence of searching for keywords and giving output by matching keywords. Since the system relies on online text to speech conversion, online query processing and online speech to text conversion hence a constant internet connection is needed to achieve all this. Speakers, once the symptoms put forward by the user have been processed, the corresponding text output is converted to speech using the text to speech converter which is the audio output is sent to the user using the speaker. First,

when the user starts the system, he/she uses a microphone to send in the input. Basically, what it does is that it takes sound input from the user and it is fed to the computer to process it further. Then, that sound input if fed to the speech to text converter, which converts audio input to text output which is recognizable by the computer and can also be processed by it. Then that text is parsed and searched for keywords. The device is built around the system of keywords where it searches the text for key words to match. And once key words are matched then it gives the relevant output. This output is in the form of text. This is then converted to speech output using a text to speech converter which involves using an optical character recognition system. OCR categorizes and identifies the text and then the text to speech engine converts it to the audio output. TTS can enable the reading of computer display information for the visually challenged person or may simply be used to augment the reading of a text message. This output is transmitted via the speakers which are connected to the audio jack of the Raspberry pi.

The system addressed here gives a general idea of how a voice system should work and we can build our medical bot on this basis. [3]

While other papers discuss about personal assistant approach and propose a way to create different cloud databases, this paper focuses on addressing the lack in database connectivity between the different cloud environment in monitoring the data in constant time intervals and to analyze data.

Rise of the Internet of Things has made several tasks easy. It helps in bedside collection of data and monitoring the same. This will surely help in monitoring the status of patients in the healthcare domain.

Considering this, a cloud based IoT system is proposed that can be implemented in different health monitoring systems. This system consists of four-protocol layers such as the physical layer, network layer, middleware layer and application layer. First, the physical layer consists of devices embedded with sensors and transmitters.

The network layer is responsible of transmitting signals from sensors to the Cloudlets whereas the Middleware layer do the work of storing the data into the cloud and makes it available to the people who are concerned. Finally, in the application layer, analytics and diagnosis process are performed. One of the main challenges faced by the authors of this paper was, having a wearable that operates with a sensor which consumes less energy.

The battery needs to be working continuously without charging and replacement. The cloudlet processing also faces challenges when it comes to having a secure framework. Secure cloud storage is very necessary as unauthorized access to some private information is very dangerous. The analytics part of the proposed method experiences three main challenges, periodic updating of the IoT devices is necessary. Machine learning algorithms must be adaptive to cooperate with the dynamic structure of the IoT system. The patient data has to be collected based on the patient condition. Since it is patient specific, it is challenging to find one particular way. The heterogeneous data from different sensors poses another challenge to the machine learning algorithms which generally use homogenous data.[4]

Model proposed in the paper: IoT-Cloud based framework for patients data collection in smart healthcare system using Raspberry Pi: In this paper, a Raspberry Pi based IoT model is presented which uses docker container instead of virtual machine for the server. Pi accepts data from sensors, and these are transferred to the cloud using server and users can access the database from the cloud.

Drawbacks: Useful when the data used is very large amount. Uses docker container which is not very relevant to our project. Intends only in sending the data to different users and nothing about the analysis of the data.[5]

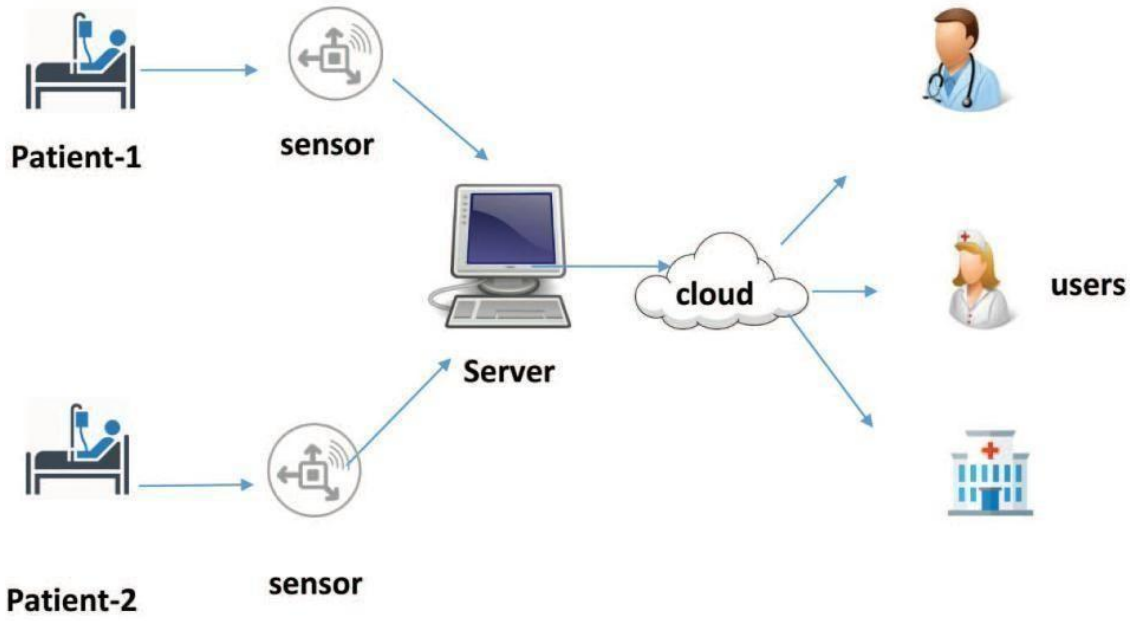

Fig 2.1 Patient data management using IOT

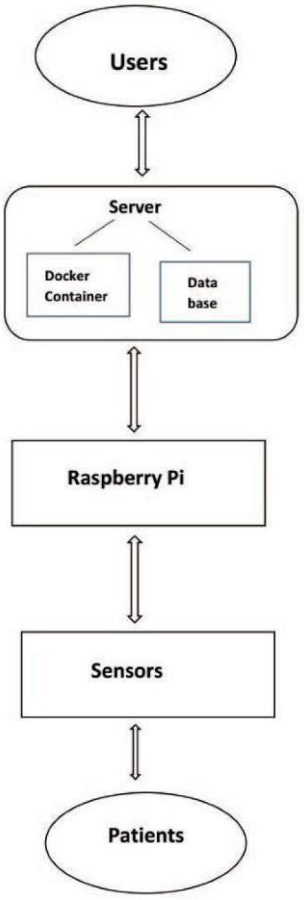

Fig 2.2 Flow diagram of the method

Sensor networks are a highly researched application area in the field of Internet of Things (IoT). A key cost and resource question in the development of IoT network sensor solutions is prototype implementation. In this study surveying of Prototyping solutions utilizing Raspberry Pi", the Raspberry Pi - a widely used single board computer-is investigated as it is one of the most commonly used prototyping devices available and is also widely used in scientific research. In this paper, we address which technologies, the usefulness and what kinds of issues arise when the prototyping of a sensor network solution is done with Raspberry Pi. The extant literature is studied by selecting papers with the systematic literature review method. Based on an extensive survey of the selected studies, we found several sensor-based implementations where Raspberry Pi has been used. In addition, this survey revealed subjects, such as e-health and education, which expanded the research topic in new ways. Further research opportunities have been identified in specifying the usefulness of various technologies with single board computers.

Sensor networks refer to distributed autonomous sensors that are used to monitor the physical environment, e.g., temperature or pressure. Sensor networks are a widely studied area. For example, the use of embedded Linux for sensor networks has been proposed and a simple model of a sensor network has been introduced earlier. There has also been research on long-range wireless sensor networks with geo-location tracking and on a low energy algorithm for a sensor network.

Single-board computers, such as Intel Galileo, Beagle Bone and Raspberry Pi, are low-cost development devices for testing or educational purposes. These are fully customizable and programmable, and have the features required implementing small and low-cost IoT devices. The Raspberry Pi is the most popular of these three in the field of research according to the keyword search. Single-board computers are the rising technology area in the development of prototypes. Often, developing a prototype is experienced as difficult and expensive due to the costs of hardware design, software design and developing as well as hardware manufacturing and building. However, by using single-board computers, these costs can be easily reduced. Ready-to-use hardware solutions already exist, such as Raspberry Pi, which have ready-to-use software with embedded Linux.

There were 15 relevant primary studies for this paper. Few of them are discussed as follows -

1. A monitoring system for detecting and preventing rodents in grain stores. The system consists of a webcam, a repeller for undesirable rodents, and Raspberry Pi with a set of sensors, which were an Ultrasonic Range Detector (URD) and a Passive Infrared sensor (PIR). The algorithm for the software was introduced. The test cases consisted of functionality tests where the hardware and software were tested. The test results showed that the observation distance was small, seven centimeters. In addition, the reliability of the system was tested. This test shows that 85 percent of system notifications were real. The 15 percent of tests that were unsuccessful were due to the connectivity of the device, data transmission, notification, and other factors such as PIR sensors being configured to generatediscretevalues.
2. Maksimovic, Vujovic and Perisic introduced IoT-based e-health systems. Their research also highlighted the economic impacts of IoT applications and especially the economic growth of healthcare applications. This research compares different applications: one is the e-health sensor shield V2.0 for Raspberry Pi and the other is a custom-made body sensor measuring system. Both enable data gathering and sending to the server application. This research does not have any special testing part, but the research extensively points out the security issues ofthe gathered data.
3. The research by Hsiao, Liang and Sung introduced a smart home system. This system uses infrared communication inside a room, Zigbee communication inside the house, and Wi-Fi communication for data transfer to the cloud. The research focus was the combination of communication types in one system. The actual sensor or controller was not introduced. The features of these communication methods were also compared. A comparison of different means of communication was made at a general level, but no specific testcases were presented.

There are a reasonable number of studies focusing on prototyping sensor networks with Raspberry Pi devices. However, most of the papers reported a single case study on the development of an interesting system. There is a clear lack of formalized approaches, methods, and tools.

Only little has been reported on the testing practices utilized, problems faced, and approaches used in the development of a sensor network or a module for it. As the testing of interconnections between the nodes in IoT networks is of utmost importance for the reliability of the system, the lack of studies isa worrying finding.

This research shows that formal software testing was used only in the minority of research. For example, often it was only mentioned that the test cases were passed. Of course, the software parts are small, especially in sensor nodes, but if the developed system has some algorithms to process the gathered data, the software may have several functionalities. These should be tested in some way. One good test is: Will the software perform its functions within acceptable time? Data validation tests were used in a minority of studies. Validation might be important when a system uses the gathered data or the results of processed data in some way.

The Raspberry Pi based prototypes are usually connected using an experiment board and soldering of connections is not common. This came up from the studies explored. During the previous studies connection faults were common and it was attempted to prevent them by soldering all possible connections. However, the articles explored in this study did not commonly handle these reliability issues. Only a few of them even mentioned these problems. Power failure is one common fault in Raspberry Pi.
Overall, the Raspberry Pi is an inexpensive, fully customizable, and programmable credit card-sized single board computer, which supports a large number of peripherals and network communication. Therefor the Raspberry Pi is suitable for small scale prototype testing and is very efficient and alsocheaper.[6]

Amongtheotherproposedmethods,thepaper–ARemoteHealthMonitoringSystemfor the Elderly Based on Smart Home Gateway, introduces a remote health monitoring system for the elderly based on smart home gateway. The proposed system of this paper consists of three parts: the smart clothing, the smart home gateway, and the health care server. The smart clothing collects the elderly’s electrocardiogram (ECG) and motion signals. The home gateway is used fordata transmission.

The health care server provides services of data storage and user information management; it is constructed on the Windows-Apache-MySQL-PHP (WAMP) platform and is tested on the Ali Cloud platform. To resolve the issues of data overload and network congestion of the home gateway, an ECG compression algorithm is applied. Evaluation of the compression algorithm shows that it has a high compression ratio and low distortion and consumes little time, which is suitable for home gateways. The proposed system has good scalability, and it is simple to operate. It has the potential to provide long-term and continuous home health monitoring services for the elderly.
Although it‘s advantages, it has some important drawbacks. It makes it mandatory for the patient to wear the smart device and doesn’t take into consideration of it getting displaced or lost. The data transmission also depends on the existence and availability of smart phones or smart devices which may not be the case always. The data transmission does not take place through thedevice itself.[7]

Having researched about the different methods available to aid the patients, some specific domains tend to serve better in building a unique solution to the problem. The following studies are on natural language processing, classification, database managementof the patient.

The paper –Medical information extraction using NaturalLanguageProcessing interpretation” is based on retrieval of medical data from narrative clinical documents using Natural Language Processing (NLP). It explains about how NLP is used to extract medical information from a narrative text by using tools such as tokenizing, noun entity recognizers, parts of speech taggers and relationship extractors. However medical text is different from normal text as it contains complex medical terminologies due to the synonym and disease names. The system that is proposed in this paper tries to understand the text relating to a list of symptoms and it will try to give out the relevant answer.
Gunjan Dhole and Dr. Nileshuke have clearly mentioned the need for using NLP in this case. According to them most information retrieval systems are rule based systems as against NLP which can extract data from free text form. Several medical extraction systems have been researched and mentioned about in the paper.

Information retrieval:

NLP stores and interprets meaning at both query and the document. A general text retrieval system would have the following flow:

1. Document processing
2. Query processing
3. Query matching
4. Ranking and sorting

The knowledge base stores all the medical documents. The system consists of two modules, document and query processing. The answer processing module will check which type of query is coming in. If it is – Definitive, it will follow the flow for definitive queries. If it is - Descriptive query, it will follow the flow for descriptive queries. Document Processing and retrieval is done from the Query processing module which gives noun entitiesas output.

Lots of research is ongoing in the field of extraction of medical data using NLP. It needs different tools as compared to the normal ones. The system described in this paper does not store the information or the medical history anywhere and the extraction techniques can be improved further. [8]

The thesis–Finding synonyms in medical texts – Creating a system for automatic synonym extraction from medical texts" focuses on bridging the gap between laypeople and medical professionals by extracting the synonyms from a corpus consisting of medical texts and to evaluate whether these words are actually related. Because, sometimes the more specific medical words are described by a more commonly used familiar word. The aim is to find synonyms and related words for different diseases. In the unigrams that are considered after tokenizing only the noun entities are retained since synonyms belong to the same part of speech (noun).

They give a stark description about the semantic relationships and weighting, which deals with giving importance to words occurring in the same context, close to the target word, giving a hint about the words true meaning. Work that has been already done in synonym extractionhas been reported.

The project is implemented in stages. Firstly, unwanted data is got rid of in the medical corpus, leaving only the necessary information.

Misspelled or unknown characters are taken as noise which should be reduced, and hence regular expressions are used which filters out irrelevant data. Next comes the part of speech filtering. They make use of a SNOMED dictionary describing diseases, words are chosen from that, which are all nouns. They only focus on unigrams. Hence filtering out all the non-noun words would lead us to the synonyms that are actually required. The pos tagger – NN is made use of. However, it is not100%accurate.

Since in large pieces of text, it is very likely that the same word will appear in different forms, NLP techniques like stemming and lemmatization are utilized. Distinguishing between words having the same root word is unwanted since it is only the meaning that is of interest to the analysis. Since Stagger has already been used for the part-of-speech tagging, lemmatizing the corpus is not more computationally costly than stemming. That is because all the lemmas have already been extracted and it is just to replace its original wordin the corpus.

Later, they go about extracting the semantically related words getting a csv file as an output consisting of the most similar words for each SNOMED term. The analysis is further done. The use of a questionnaire is stated to interact with the user to narrow down to disease they are conveying. A human evaluation on a few selected members is performed and the results and the statistics are explainedin the paper.

The limitation of this system is that it only handles unigrams which could decrease the overall performance of the system. Bigrams or further ngrams would also describe the adjectivesforaparticularsymptom,like–runnynose'or–severepain'.Thisemphasizestheneedfora model that can manage n-grams larger than unigrams. The noise removal is also not very efficient in retaining only the noun parts of the user input. When replacing synonyms in texts, it is imperative that no information gets lost, especially in a medical context where correct information could be the difference between life and death. Dealing with this sensitive and personal information is difficult and entails many ethical questions that have to be considered. Hence there is much room for improvement on the aforementionedsystem.[9]

The paper –Analysis of Various Decision Tree Algorithms forClassificationinDataMining'discusses on what and whydecision treesareusedinDataMining and some of the most popular and widely used algorithms used to create decision trees, namely: ID3(Iterative Dichotomiser), C4.5,CART,RandomForest.

1. ID3:

a. This method uses greedy algorithm and is based on Occam's Razor. It creates a tree by searching the entire dataset. It creates a very fastand short tree.

b. It can lead to over-fitting or under-fitting of tree. It is time consuming and can be computationally expensive.

2. C4.5:

a. It is an extension of ID3 algorithm. It is very easy to implement. It can be used for both continuousand discrete values.

b. Itdoesn_tworkwellonsmalldatasetswhichcanprovebetroublesomeforour project.

3. CART:

a. This method uses binary splitting of attributes. It works well on continuous and discrete data. It is fast and can handle missing values well. It works well on small datasets.

b. It can be easilyinterpreted by humans.

c. It splits at only one variable. It can createunstable trees sometimes.

4. Random forest:

a. It is a collection of simple tree predictors. It is able to recognize anomalies and outliers. It is one of the most accurate methods and gives the important or major attributesaffecting the classification.

b. It can easily lead to over-fitting and is hard tointerpret by humans

Based on the advantages and drawbacks by the above methods, **Random Forest Classifier** was shown to be a better match for our project's disease classification. [10]

The paper–Hospital patient database management systempresents several examples and reasons for choosing a computerized medical data recording system over traditional methods.

It aids the international patients: Few patients travel from one country to another for specialized treatments, in such situations, having a paper-based recording system is time consuming. Every second becomes precious when it comes to a person's life. This can be avoided using acomputerized database management.

Storage of physical records is cumbersome: Physical records usually requires significant amount of space to store to them, when physical records are no longer maintained, the large amount of storage space are no longer required paper film and other expensive physical media usage can be reduce by electronic record storage. Paper records are stored in different location, furthermore, collecting and transporting them to a single location for review by a health care provideris time consuming.

Computerized database system helps is reducing waste: The paper presents a case of how a drug company reduced the drug wastage by implementing the computerized database management system which kept account of the drugs that are present and have fallen shortage inthe drug inventory.

This paper presents enough proofs and information about why we need a computerized medical record management system. [11]

The paper - Hospital Database Management System presents the difficulties ofmaintaining a computerized medical record in traditional applications like Microsoft excel and Google docs. It also presents few main barriers present in implementing ahealthcare database.

Healthcare database differs from a traditional database: Health care database includes many more information and fields that have to be created and managed.

Example: patient, disease, symptoms, medication, sensor values etc.

Interconnected database has to be created: To provide each patient with related help and aid, there has to be a database that is interconnected. Hence relational database is necessary.

Different departments and medical vocabulary: A health care sector includes different departments within it. All these departments have to be addressed and relatively the data has to be stored. Medical vocabulary is complex and that also has to be addressed by the database.

Prediction based on history: The paper talks about Stanford university research where a predictive diagnosis is presented for patients based on previous records. Thus, creating a database system helps in statistical analysis of the diseases and prediction.

Among all the database structuring methods available, most software professional are eager to opt for relational database as the technology has well trained professional and much expertise & user-friendly.

Having studied the advantages and disadvantages of all the papers discussed above, the proposed method absorbs the necessary and useful elements from each of them. Doing so, our device tries to overcome the cons with preferableimprovements.[12]

CHAPTER 3

PROPOSED METHOD

Chapter – 3

PROPOSED METHOD

3.1Input Interface -NLP

The basic ideology of the input part is to implement a voice system that can take in the symptoms the user is facing and to incorporate it into the classification algorithm and output the estimated prognosis clearly.

Using Raspberry Pi as a main hardware to implement this model which works on the primary input of a user's voice as an input, it is used to convert into text using a speech to text engine. The text hence produced is used for processing and fetching relevant information. When the information is fetched, it is then converted to speech using text to speech conversion and the relevant output to the user is given. Additionally, some extra modules are also implemented which works on the conceptof keyword matching.

Any voice system would need three basic components which are speech to text converter, query processor and a text to speech converter. Voice has been a very integral part of communication nowadays. Since, it is faster to process sound and voices than to process written text, hence voice command systems are omnipresent in computer devices. Text to speech conversion is the process of converting a machine recognized text into any language which could be identified by a speaker when the text is read out loud. It is a two-step processes which is divided into front end and back end. First part is responsible for converting numbers and abbreviations to a written word format. This is also referred to as normalization of text. Second part involves the signal to be processed intoan understandable one.

Speech Recognition is the ability of a machine to understand words and sentences spoken in any language. These words or sentences are then converted to a format that could be understood bythe machine.

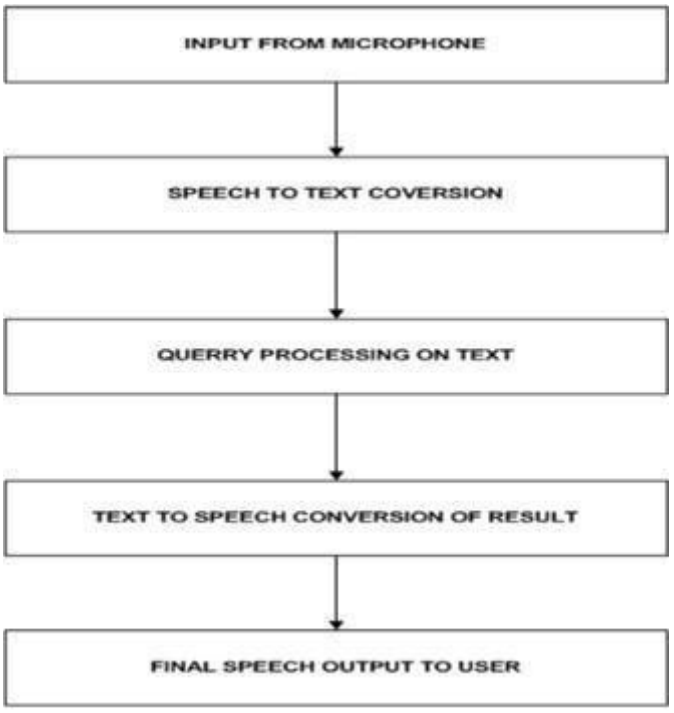

Fig 3.1 Input Interface flow

Flow:

The user input is the query input into the module and important entities will be extracted from the query and proper answers will be extracted. Answer matching will try to find the best possible answer, in this case the disease out ofthe given set.

Data extraction and processing will involve management of the database which could contain text relating to the diseases. With NLP, processing involves section splitting, tokenization and relationship extraction. The query is also processed using NLP and will extract all the important relationships from the query and the closest answer or the disease will be predicted. Hence if the user inputs a set of symptoms the system should be able to predict the closest disease with a good recall. The dataset should be maintained with all the corpus ofthe disease names.

Input:

User Input Interface: The user input interface accepts the query from the user. The user interface will accept this query and give output as the disease related to these symptoms. And if the query is asking information about the disease it will give the disease information.

They discuss about two kinds of queries:

1) Definitive – Queries asking informationabout any disease

Queries like – What is lung cancer What do you mean by lung cancer Meaning of lung cancer

2) Descriptive – Queries describing about the symptomsof the patients

Queries like – I have coughing, coughing up blood, wheezing or shortness of breath. What is my disease.

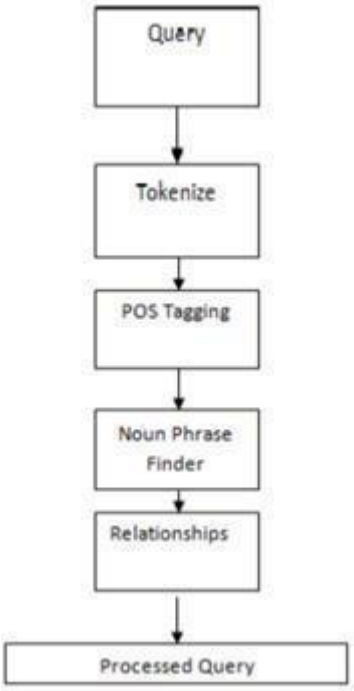

Fig 3.2 Query Processing

The query processing consists of:

Spell Check: If the words from the query do not exist in the English dictionary, some suggestions are given by interchanging the characters, replacing characters and inserting new characters into the word. Now from this list of suggestions the word with minimum edit distance will be chosen.

Input: What is lung cancer? Output: What is lung cancer?

Capitalization: For the correct syntax to process the query using NLTK, proper noun words are capitalized. This is being done with the help of corpus which contains the disease names. If the Ngram from the query matches with the disease name, then capitalize the ngram and change the query according to that.

Input: What is lung cancer? Output: What is Lung Cancer?

Tokenization and Pos Tagging: Tokenization tries to get the individual entity from the query and pos tagging tries to part of speech tag each word with the same tags.

Input: What is Lung Cancer

Tokenization: [“what”, “is”,"Lung","Cancer"]

Postagging:[(–what","WH"),(–is","VBZ"),(–Lung","NNP"),(–Cancer","NNP)

Noun Entity Recognition: It is done either with the NLTK noun entity recognition or chunking different pos tags together. Input: What is Lung Cancer Output: [–Lung Cancer”]

3.1 Input Module - GUI

GUIstands for graphical user interface, and it is pronounced like –gooey". Generally, programs take in input from the users and output displayed on screen. Mostly on command prompt. Adding GUI to the program lets user interact with the program using buttons, menus, text boxes and other familiar user interface features.

To create the GUI, python module is necessary. This is compatible with the raspberry pi too. Using the general construction procedure of a GUI, the UI is built, the output from the speech to text convertermodule is interfaced with the UI. Thus, helping the project interact with the user conveniently. Tkinter aids the development of GUI. It supports all platforms; it has several widgets to offer.

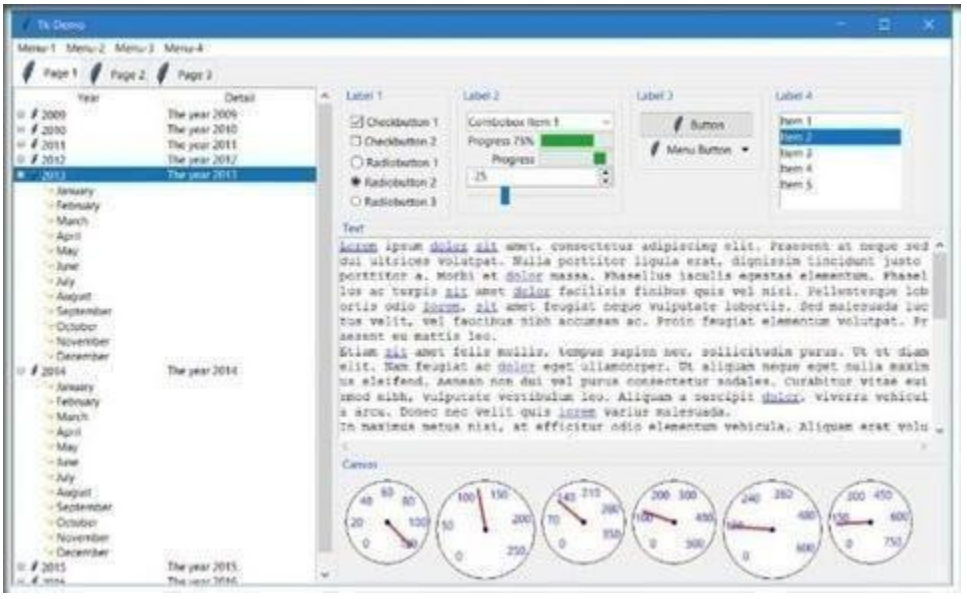

Fig 3.3 GUI widget options on Tkinter

3.2 Classification of Data using RandomForest Classifier

Bhaskar N Patel et al. sought to study the different Data classification means categorization of data according to rules of the different rules. The aim of the proposed research was to extract a kind of –structure from asample of objects. The paper discussed one of the most widely used supervised classification techniques, which is the decision tree and performed its own Decision Tree evaluate strength of own classification with Performance analysis and Results analysis.

To rephrase it, to learn a concise representation of these data. Present research performed over the classification algorithm learns from the training set and builds a model and that model is used to classify new objects.

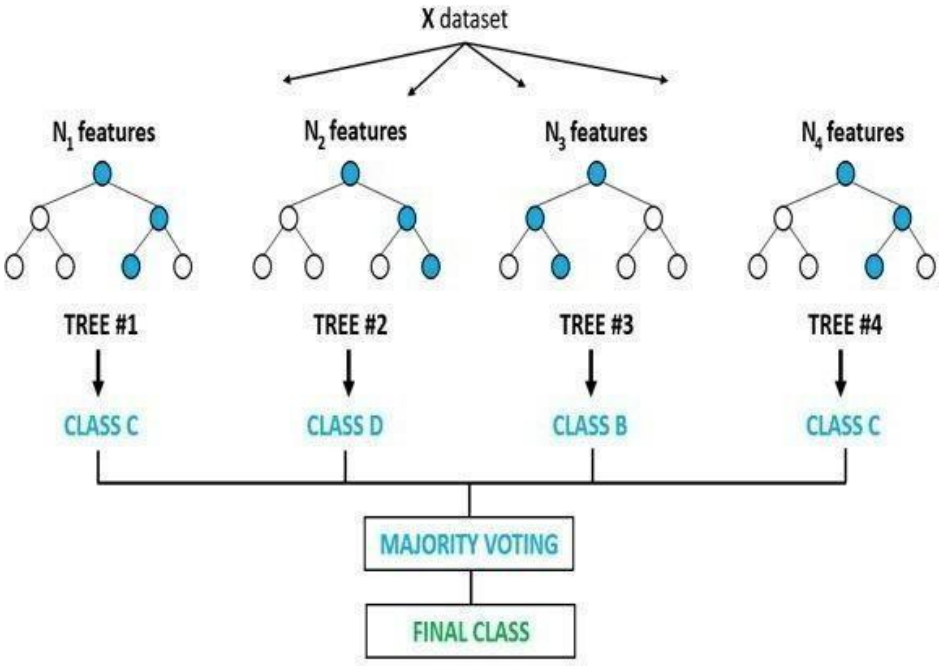

Fig 3.4 Random Forest Classifier

As per the implementation of this project, data classification with random forest classifier was found to be easy when compared to other methods.

The fundamental concept behind random forest is a simple but powerful one — the wisdom of crowds.

The random forest is a model made up of many decision trees. Rather than just simply averaging the prediction of trees (which we could call a –forest),this model uses two key concepts that gives it the name *random*:

- Random sampling of training data pointswhen building trees.
- Random subsets of features considered whensplitting nodes.

Random Forest is an extension over bagging. Each classifier in the ensemble is a decision tree classifier and is generated using a random selection of attributes at each node to determine the split. During classification, each tree votes and the most popular class is returned.

Key concepts to understand about random forest

- **Decision tree**: an intuitive model that makes decisions based on a sequence of questions asked about feature values. Has low bias and high variance leading to over-fittingthe training data.

- **GiniImpurity**: a measure that the decision tree tries to minimize when splitting each node. Represents the probability that a randomly selected sample from a node will be incorrectly classified according to the distribution of samples in the node.
- **Random subsets of features**: selecting a random set of the features when considering splits for each node in a decision tree.
- **Bias-variance tradeoff**: a core issue in machine learning describing the balance between a model with high flexibility (high variance) that learns the training data very well at the cost of not being able to generalize to new data , and an inflexible model (high bias) that cannot learn the training data. A random forest reduces the variance of a single decision tree leading to better predictionson new data.

Bagging (Bootstrap Aggregation)

Decisions trees are very sensitive to the data they are trained on, small changes to the training set can result in significantly different tree structures. Random forest takes advantage of this by allowing each individual tree to randomly sample from the dataset with replacement, resulting in different trees. This process is known as bagging.

Features of Random Forests

- It is unexcelled in accuracyamong current algorithms.
- It runs efficiently onlarge databases.
- It can handle thousands of input variableswithout variable deletion.
- It gives estimates of what variables are importantin the classification.
- It generates an internal unbiased estimate of the generalization error as the forest building progresses.
- It has an effective method for estimating missing data and maintains accuracy when a large proportion of thedata remissing.
- It has methods for balancing error in class populationunbalanced datasets.
- Generated forests can be saved for future use onother data.
- Prototypes are computed that give information about the relation between the variablesand the classification.
- It computes proximities between pairs of cases that can be used in clustering, locating outliers or (by scaling) give interesting viewsof the data.
- The capabilities of the above can be extended to unlabeled data, leading to unsupervised clustering, data views and outlier detection.
- It offers an experimental method for detecting variableinteractions.

Implementation steps of Random Forest –

1. Multiple subsets are created from the original data set, selecting observations with replacement.
2. A subset of features is selected randomly and whichever feature gives the best split is used to splitthe node iteratively.
3. The tree is grownto the largest.

4. Repeat the above steps and prediction is given based on the aggregation of predictions from nnumber of trees.

Each tree is grown as follows:

i. If the number of cases in the training set is N, sample N cases at random - but with replacement, from the original data. This sample will be the training set forgrowing the tree.

ii. If there are M input variables, a number m<<M is specified such that at each node, m variables are selected at random out of the M and the best split on these m is used to split the node. The value of m is held constant during the forest growing.

iii. Each tree is grown to the largest extent possible. Thereis no pruning.

In random forests, there is no need for cross-validation or a separate test set to get an unbiased estimate of the test set error. It is estimated internally, during the run itself.

The random forest combines hundreds or thousands of decision trees, trains each one on a slightly different set of the observations, splitting nodes in each tree considering a limited number of the features. The final predictions of the random forest are made by averaging the predictions of each individual tree.

3.3 Hardware

Examination of vital signs such as heart rate and body temperature are critical to be done by every human being to know the condition of the new illness suffered and also the physical health. The heart is the center of the human circulatory system and it is in record that 35% or 1.8 million Indonesians in 2014 died of a heart attack. Body temperature changes affect the body condition that can cause infection, inflammation, and stress. Patient healthcare factors, especially heart rate examination and body temperature are fundamental parameters for medical personnel in diagnosing disease, maintaining the safety of the soul, and the patient'sphysical condition.[16]

The design of the heart rate monitoring system and body temperature includes both hardware and software. Pulse heartbeat 3554 sensor serves to retrieve the patient's heartbeat data by inserting the thumb to the end of the sensor. Any detected sensor (patient) data will be received by Raspberry Pi and perform data processing in beats per minute (bpm). Results of sensor data processed in Raspberry Pi module will be displayed on the LCD interfaced to it and also the results are stored in the Google Firebase Cloud.

DS1820 temperature sensor serves to retrieve body temperature data by holding the tip of the sensor. The result of body temperature detection is received by raspberry pi and processed through conversion from temperature to voltage. The resulting voltage changes will adjust to the condition of the body temperature of the patient at that time. The result of a translation made by raspberry pi will be displayed on LCD. All patient data stored on raspberry pi can be called or viewed on the portable or personal computer.

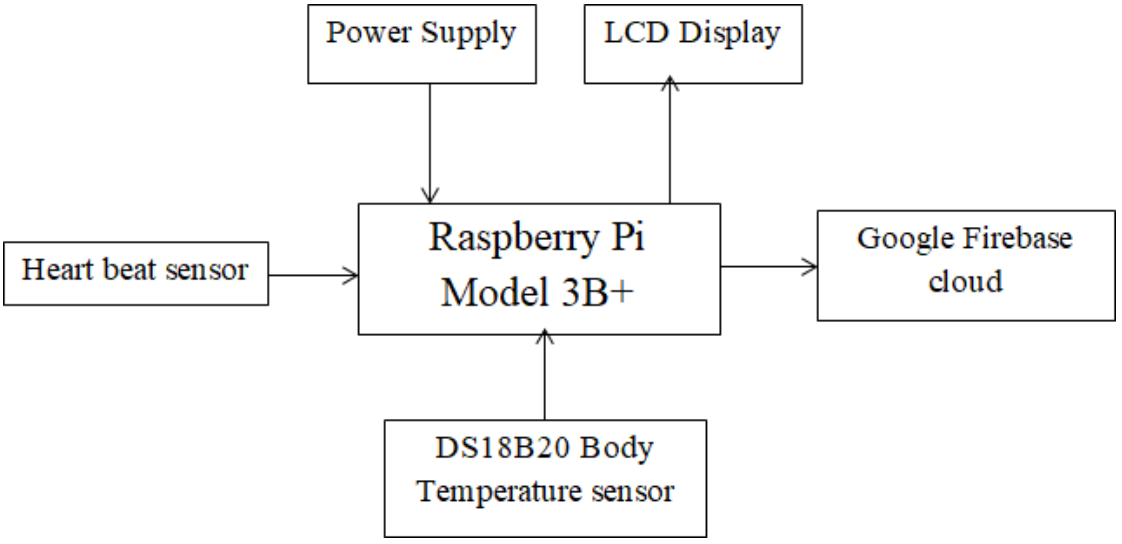

Fig 3.5 Proposed block diagram for sensor data measurement

The software stage consists of the Integrated Development Environment (IDE) application program used in the Arduino module with the C ++ programming language. IDE application works to write, display, and upload program code. Results of program data created in the text editor (sketch) stored in files with extension (.ino). This design

begins with the creation of pulse heartbeat 3554 sensor program on Arduino Uno in the form of input voltage from the sensor. Sensor data serial communication with raspberry pi. If pulse heartbeat 3554 sensor is readable then it will be sent to the raspberry pi, but if the sensor is not legible then go back to the initial reading. Raspberry Pi will read Arduino serial data simultaneously with temperature sensor readings. Temperature sensor data readings are sent directly to raspberries in the form of input voltage converted from the library in the way of Fahrenheit then turned to Celsius. Furthermore, the temperature sensor data and heart rate will store in the database. Sensor data is displayed on the LCD and can check in portable via an internet connection.

Testing the tool performed before taking the data heartbeat. This test aims to ensure the sensor works well in all the conditions. The result of the pulse heartbeat 3554 sensor testing shows that the output value read on voltmeters of 3.27 volts is not much different from the voltage value on the pulsing heartbeat 3554 sensors of 3.3 volts. The next test performed on the sensor indicator by observing the red led indicator. The red led lights give a sign that the sensor is working correctly and if the red led does not light it indicates that the sensor is not working. This problem caused by the placement of the sensor position on the finger is not appropriate, and the sensor connectors are not connected properly.

DS18B20 temperature sensor testing is done by holding the end of the sensor or pair the sensor on the armpit for two or three minutes. Armpit is a body part that can use as a basis in the sensor readings. The readings of the body temperature sensor on the LCD in the form of

Celsius reading indicates that the sensor works well. Furthermore, to obtain accuracy testing of temperature sensor data, a digital thermometer was used as a comparison in getting the error rate. The success of this test can see in LCD or portable readings. The measurement results displayed on the LCD is an indicator that the sensor DS18B20 works appropriately.

Before the patient data storage process is done, filling of the patient data on the health monitoring application menu should be done in prior. Storing patient data consists of name, age, address and phone number. After the data input process is over, automatically patient data is displayed. The health monitoring information presented consists of the

time, heart rate, body temperature, bpm signal that works in real time. The results of patient data and sensor data readings are stored on the raspberry pi module. This module works to store and transmit patient data to portable or smartphone devices.

Distance range of the proposed tool transmission reaches 67 meters for open space conditions without obstructions and 13 meters for enclosed spaces with obstructions. The proposed tool has an error rate in detecting the heartbeat of 2.22% and body temperature of 0.27%.

3.4 Management and Analysis of patient data

Medical record of a patient holds very important information which can be used for several purposes. Typically, people use papers and handwritten documents to keep track of a person's health conditions. This becomes cumbersome for analysis and takes long time to gather one set of information. Thus, creating a database is a time saving and very useful task.

Proposed system

Google Firebase is a Google-backed application development software that enables developers to develop iOS, Android and Web apps. Firebase provides tools for tracking analytics, reporting and fixing app crashes, creating marketing and product experiment.

Firebase

Firebase is Backend as a Service (BaaS) by Google. Firebase provides many tools to develop high-quality apps, grow user base, and earn more money.

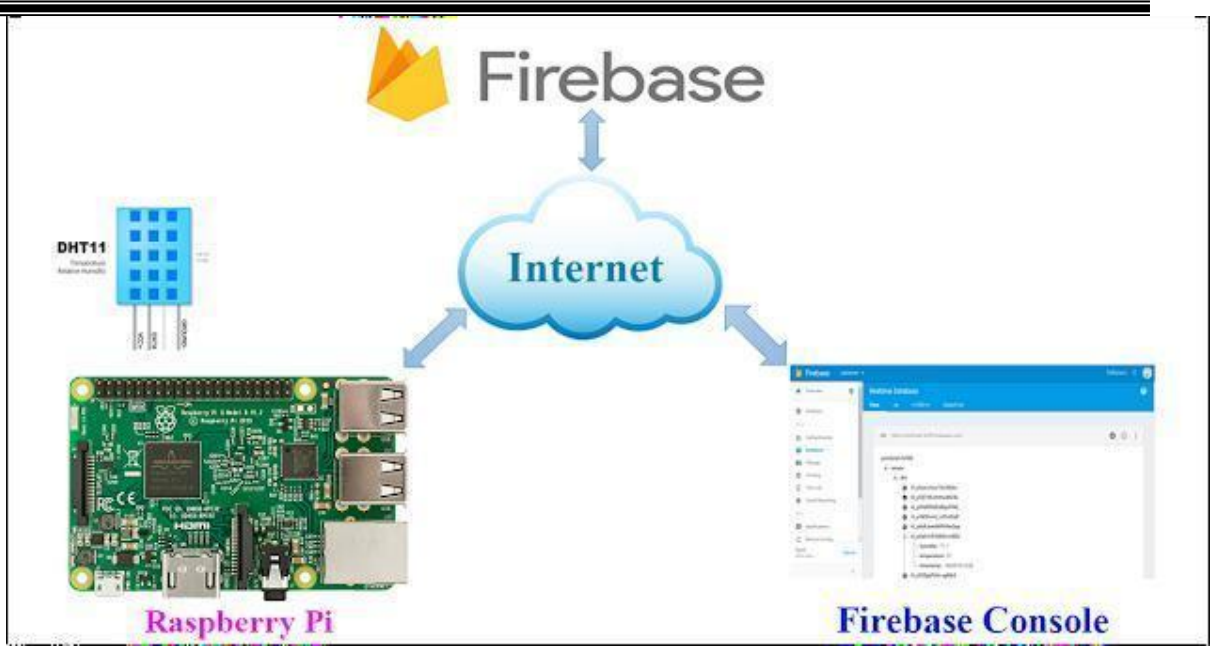

Fig 3.6Raspberry Pi connected to Google Firebase

Real Time Database

The Firebase Realtime Database is a cloud-hosted NoSQL database that lets you store and sync data between your users in real time. Updated data syncs across connected devices in milliseconds, and data remains available if your app goes offline, providing a great user experience regardless of network connectivity.

Advantages: It is free and easy to use, also provides analytical aid, it can also be used to build apps to provide the analytical details about the data.

3.5 Blockdiagram and Description

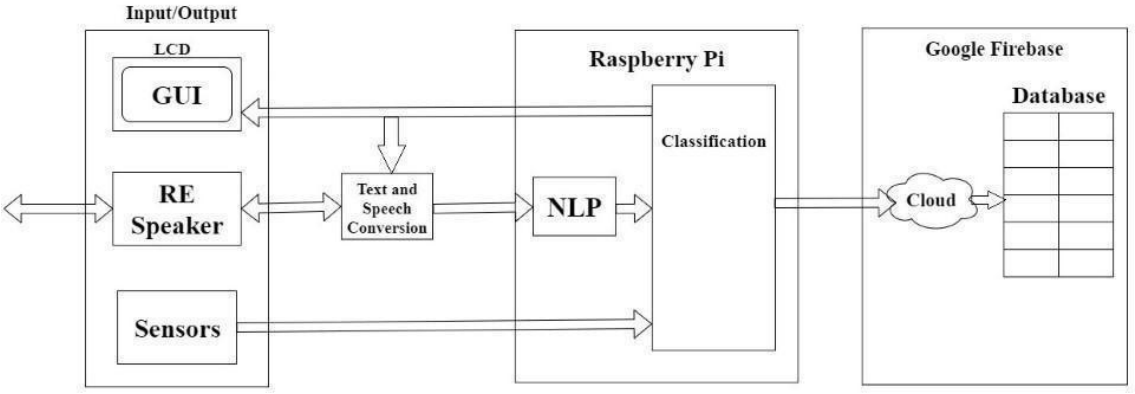

Fig 3.7 Proposed blockdiagram

The flow of the project is conceptualized in the block diagram as shown. The whole system is embedded on a Raspberry Pi Model 3B+ (RPi) board, which acts as the heart of the device. Starting off with user interface part, the user interacts with the device via speech input which will be captured by a RE speaker that is mounted on the RPi. Since the RPi Model 3B+ doesn't inherently possess a microphone for capturing audio input, we use RE Speaker for capturing the speech input. The audio is converted into the text format using speech to text convertor and is fed to the natural language processing software. Here the text is parsed, and the required keywords are extracted and matched with the dataset of disease-symptoms that is stored a priori in the RPi. The matched keywords are then passed to the classification algorithm that takes place using a tree which filters out the closely matching disease through further clarification as per the user inputs. The clarification and the predicted diseases are all displayed on an LCD screen through Graphical User Interface (GUI). Additionally, sensors are interfaced to the RPi in order to capture vital signs like temperature and pulse rate, which are transferred into the classification block. Having done the prediction of the disease along with the remedial information alongside, all the interaction data that includes the prognosis, the symptoms that led to the prediction and other information are all stored in a Google firebase database which is a Baas provided by Google with many tools. This stored information can be used in the future for statistical analysisof any kind.

3.6 System Requirements

Hardware Requirements

- Raspberry Pi3ModelB+
- 3.2-inchRaspberry Pi Display
- Respeaker2-MicsPiHAT
- DS18B20 WaterproofTemperature sensor probe
- HeartRate Pulse sensor
- MCP3008 ADC

Software Requirements

- ProgrammingLanguage:Python3.7
- Operating System: macOS / Windows8/1064-bitversion
- Raspbian OS forRaspberry Pi operation
- Cloud Database: Google Firebase console
- GUIbuilding software: tkinker

CHAPTER 4

IMPLEMENTATION

Chapter – 4

IMPLEMENTATION

4.1 SOFTWAREIMPLEMENTATION

4.1.1 USERINTERFACE–NLP

For a device to accept voice input and understand it, certain steps have to be followed to parse the input. The proposed method includes the following steps:

1 Speech synthesis
2 Modifying the dataset
3 Interpreting the input
4 Review

We have made use of NLTK package to perform the tasks, due to its easy to use interfaces such as WordNet, and a suite of text processing libraries for tokenization, stemming, tagging, parsing and so on, all of which shall be explained below. It is a free open-source package. Pandas library is a fast, flexible and a powerful tool that can be used for reading datasets in our project.

The block diagram of the proposed method is as shown in Fig 4.1.

A.SPEECH SYNTHESIS:

Starting off with the process, the prime component is the speech. It is random and cannot be directly used. Hence, an analog to digital conversion is done to digitize it to convert it from the raw physical sound to an electrical form with the help of a microphone. Once this is done, there are several models that transcribe the audio into the textual form. There are several packages/ APIs to perform this.

We have made use of one of the popular ones, that is, Google Text to Speech API (gTTs). It is a convenient tool. The Speech Recognition library retrieves the audio input and is very flexible. The Recognizer class of Speech Recognition recognizes the speech and has many methods that it makes use of, using different APIs, like gTTs. It requires an internet connection butis quite facile.

We would also need an additional tool for the reverse that is text to speech. The pyttsx module does text to speech conversion and also works offline.

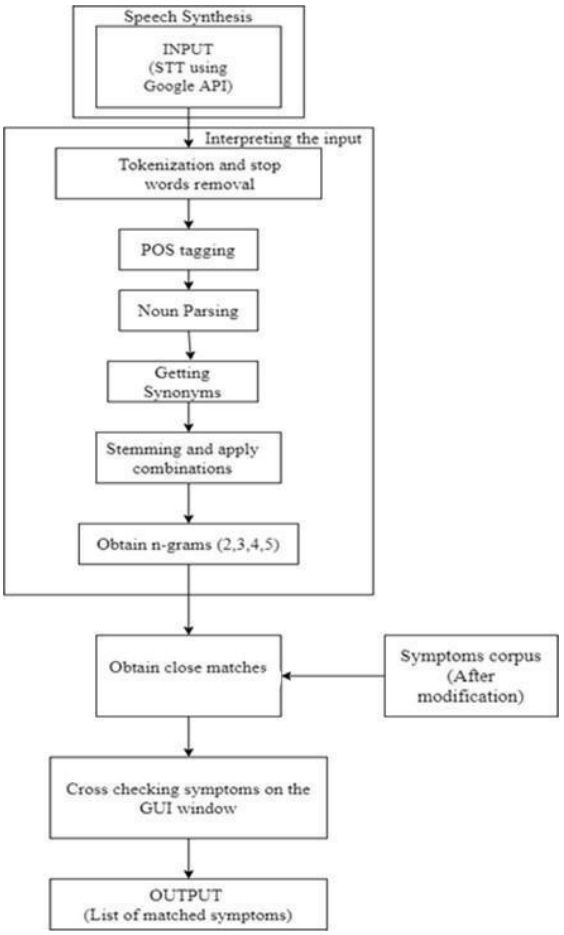

Fig.4.1 Block diagram of the proposed system

B.MODIFYING THE DATASET:

Any symptom checker would have a dataset of a list of diseases and symptoms as a basis for prognosis. We start out by considering a small dataset, which could be a .csv file and is modified slightly. Since the user input will not be in the exact terms as in the symptoms dataset, we take the root words of the symptoms. Hence, if the user's words have the same root word(s) of that of the dataset, it matches. Taking the root word is done by processes like stemming or lemmatization. They are collectively called as text normalization processes of NLP.

Example: Thewords "suffering" , "suffered","suffers" havethesamerootwordi.e. "suffer".

Stemming deals with reducing the word to its root, even if the root isn't a meaningful word. "Porter Stemmer" of the "nltk.stem" package is one of the oldest and popular one that is in use. It is known for its simplicity and speed.

Lemmatization, however, reduces the word to a meaningful root word which is called as lemma.PythonNLTKprovides–WordNetLemmatizer'thatuses–WordNetDatabase'to lookup lemmas of words. This modified dataset will be kept as a reference for further steps.

C.INTERPRETING THE INPUT:

Here comes the interesting part! Making sense out of the input and obtain the relevant information (i.e. symptoms) that can be made use for classification. The raw input is a string from the speech synthesis. All the actions are performed on this string.

One of the important steps of any NLP method is text tokenization. It involves splitting words or sentences from a body of text into entities. It is carried out by "word_tokenize" module.

Example:Input– "Iamfacing severepain in the chest"

Output–["I", "am", "facing", "severe", "pain", "in", "the", "chest"]

Stop words are those words that are insignificant and are usually filtered out because they contain a vast amount of unnecessary information. The common ones used in englishlanguage are "as", "that", "of", and "soon".The "stopwords" corpusofnltkgivesthe collection of such words. Hence, they are removed from the tokenizedlist.

Next, we do what is called as part of speech (pos) tagging. It is also called as grammatical tagging. The goal of a pos tagger is to apply a linguistic tag or information to the tokens. Since we're interested in only extracting the symptom part of the text, which are commonly forms of nouns or adverbs (mostly), we filter in only such tagged words. Our process becomes more efficient step by step.

Some common WordNet methods are applied like that of –lexeme", which gives the tense forms of a particular word.

Another important factor to be considered is a lexical relation called synonyms. Since the limited dataset obviously cannot cover all the similar meaning words that it contains, we need to simply find the same for the input and compare it with the dataset. The "synset'packageprovidesalistofsynonymwords.

Example:Input--ache'

Output - {'ache', 'aching', 'hurt', 'languish', 'pine', 'smart', 'suffer', 'yearn', 'yen'}

We will also be considering two, three, four or n-tokens together and apply all of the above-mentioned methods, because the symptoms could also be a group of words. They are called as n-grams and n could be any positive integer. The commonly used as bi and tri-grams. They are compared against the dataset.

To be on the safer side, we will use a comparing algorithm on the input n-grams with the dataset symptoms, to get the closest matching words. Such an algorithm is made use in the "fuzzywuzzy"module of Python. It is based on the Levenshtein distance to calculate the difference between the given two texts or strings. It has a value that is used as a threshold to compare and provides all the closely related words, completing our text parsing.

In order to implement the fuzzy search based on the phonetics of a word which can prove to be very efficient compared to the previous methods, we make use of two phonetic algorithms, i.e. Soundex and Metaphone.

Soundex is an algorithm and is based on how two words are depending on their English pronunciation. It converts the input string into a 4 character long alphanumeric output which can be compared with the Soundex value computed for the other string. The same key values are the same for twosimilar sounding words.

Whereas Metaphone, which is a refinement on the Soundex algorithm, has a wider set of English pronunciation rules allowing for various lengths of keys.

For example: The Soundex and Metaphone values of the following two symptoms are:

A= ‘musclewasting’ B = ‘musclepaining’

Soundex(A)= ‘M242’

Soundex(B)=‘M241’Metaphon

e(A)= ‘MSKLWSTNK’ Metaphone(B)=‘MSKLPNNK’

From the above example, when comparison is done on the basis of how similar sounding two words are, we note that the Soundex values differ by 1 whereas the Metaphone values of the second word are not very similar, indicating that Metaphone gives more accurate results.

Several other methods can also be applied to make the output more accurate; however, the efficiency and the time taken must be kept in mind. The output from each and every stage of all the NLP processes is compared with the dataset symptom and any closed match is recorded inan output list.

D.REVIEW:

Since we always must err on the right side, the recorded output list of symptoms is cross-checked with the user. A yes/no question-answer type interface is made for each symptom and recorded again in a list. If the output is an empty list, the user is asked if they would want to start over or exit. Finally, the verified output list is the result of the NLP performed on the input and is passed on to further classification processes.

Additionally, a generaluserinterface(GUI)isbuiltwiththehelpof "tkinter" Python library. The NLP will be running in the backend while a GUI is present in the frontend providing the user a visual experience along with audio updates.

4.1.2 CLASSIFIER –RANDOMFORESTMODEL

For implementation of the classifier, first, a model based on simple binary Decision Tree was built. It was a model with a node matrix for the training and testing of the Data Frame. Each symptom present against its disease was marked as –1"else as–0'. Although the model was built there were issues faced during entering of the symptoms as well as the predictions. Due to the above stated problems, the model was dropped.

The second model that we tried was based on Random Forest Classifications. It used Label Encoder for pre-processing and fitting the transform of the target variable to achieve classification. It took in symptoms as either – '1' or '0' as 'Yes' or "No' respectively. Although the model was promising in results, the process of entering inputs was tedious and long. Further, it also was not compatible with the input methods used by the NLP section of our project and our proposed hardware. Thus, this version of the classifier was also dropped.

Due to the promising results shown by the second model, it was decided for us to continue using Random Forest Classifier as our classifier. Elements of the previous model such as Label Encoder for pre-processing, fit transform of the target variable and consideration of target variable probability was kept intact. To achieve a model based on our requirements and proposed idea many changes were made.

The working of model of classifier to be used will be explained in the further section.

A. WORKING OF THERANDOMFORESTCLASSIFIER

Before any classification process, the data or Data Frame must be prepped and pre-processed as standard procedure.

This is achieved by pre-processing package imported from sklearn. The method known as Label Encoder from the pre-processing package is used for data prepping. The Label Encoder is used to label encode and convert the data in machine readable form.

Itassignsauniquenumber(startingfrom0)toeachclassofdata.Here,theencodeddataisthetarget variableorthe "Prognosis" and used infitting and transformfittingthemodel. For the Random Forest implementation, the RandomForestClassifierpackageisimportedfromsklearn.Thismodelmakesuseoftransformfittingbythepackagetofit the data. Thetransformed data being thetarget variable,–Prognosis'.

The number of trees is an important factor in forest building. From testing, it was observed that number of trees i.e. n_estimators = 10 gives the better results. Thus, the number of trees in our model is 10 and is given as input attribute to the Random Forest Classifier function used.

From the package sklearn, metrics and classification_report is also imported.

In the model, four methods of the package Random Forest Classifier is used namely: fit, fit_transform, predict_proba and classes.

i. **fit**: It is used to fit the data into the required model, here, as a random forest classifier. It builds a forest of trees. The input being the DataFrame albeit the target variable,–Prognosis'.

ii. **fit_transform**: It is used to transform data and then fit into our required model. The input for this method is the target variable,–Prognosis'.

iii. **predict_proba**: It is used to compute the probability of classes of the DataFrame. Here, the classes as the diseases or–Prognosis'.The predicted class probabilities of an input sample are computed as the mean predicted class probabilities of the trees in the forest. The class probability of a single tree is the fraction of samples of the same class in a leaf.

iv. **classes_**: It gives the classes labels (for a single output problem), or a list of arrays of class labels (for a multi-output problem).

B. STEPSINCLASSIFICATION

1. Using the symptoms given by the user initially, the code creates a list of the symptoms covered and a new DataFrame. The symptoms list created appends each symptom one at a time after checking if such a symptom exists in our training set of symptoms. The new DataFramecalled–df_copy'iscreatedsuchthatitcontainsonlycolumns of the enteredsymptoms.
2. Further,usingthis–df_copy'probability of possible diseasesisdetermined. It checkseach disease that has one or more symptoms of the entered symptoms and predicts the probability of the new list orclassofdiseases.
3. On calculation, a new list of the possible diseases is created. As an additional step to narrow down on the prediction, the disease with least or minimum probability is removed. At this stage, to get the disease names in user readable form inverse transform is taken to cancel the transform done at the beginningoftheprediction.
4. On the output side, the symptoms covered, the possible diseases, probability of possible diseases and the remaining symptoms aredisplayedaslists.
5. The probabilities of the diseases are checked, if one of them is above 0.7 then the suspected disease is returned and breaksfrom the loop.
6. Else, if none of the probabilities are above 0.7, there exists a loop to allow the user to add more symptoms. This step is prompted by remaining symptoms of the possible diseasesin the list.
7. Thereisalsoaprovisiontonotaddanysymptomsbygivinginputas–None'.
8. This loop continues for 2 iterations. If within 2 rounds, a disease has probability above 0.7, it is returned as suspected disease andloopisbroken.
9. Else if by the end of 2 iterations, a singular suspected disease is not narrowed down and if none of the possible diseases have probability above 0.7, then theloopisbroken.

4.2 HARDWAREIMPLEMENTATION

4.2.1 DS18B20 WATERPROOFTEMPERATURESENSOR

DS1820 temperature sensor serves to retrieve body temperature data by holding the tip of the sensor. The result of body temperature detection is received by raspberry pi and processed through conversion from temperature to voltage. The resulting voltage changes will adjust to the condition of the body temperature of the patient at that time. The result of a translation made by raspberry pi will be displayed on LCD. All patient data stored on raspberry pi can be called or viewed on the portable or personal computer.

DS18B20 temperature sensor testing is done by holding the end of the sensor or pair the sensor on the armpit for two or three minutes. Armpit is a body part that can use as a basis in the sensor readings. The readings of the body temperature sensor on the LCD in the form of Celsius reading indicate that the sensorworks well.

The DS18B20 sensor interface with the Raspberry Pi is simple as shown in the figure below, making it a highly portable device. Connection complexity is less and the sensor is easy to handle, thereby ensuring optimum comfort when it is used for measuringthe body temperature.

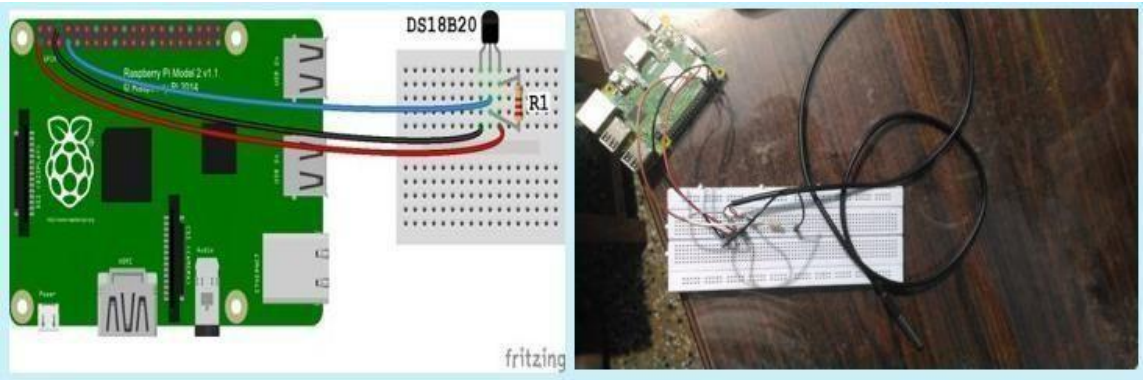

Fig 4.2: 2D and 3D view of the DS18B20 temp. sensor interface with the Raspberry Pi

CHAPTER 5

RESULTS AND ANALYSIS

Chapter - 5
RESULTS AND ANALYSIS

5.1 SOFTWARE

5.1.1 NLP

The proposed methodology was implemented using Python (version 3.7.3) and user interface was developed using tkinter package and the results obtained are presented in this section.

Example 1:

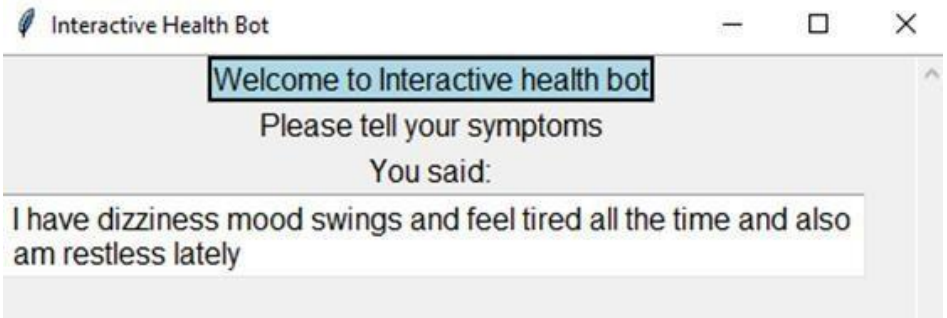

Fig 5.1Recording the user input

The voice input is taken into the system. Fig.5.1 shows an example of an input sentence describing the symptoms of the user.

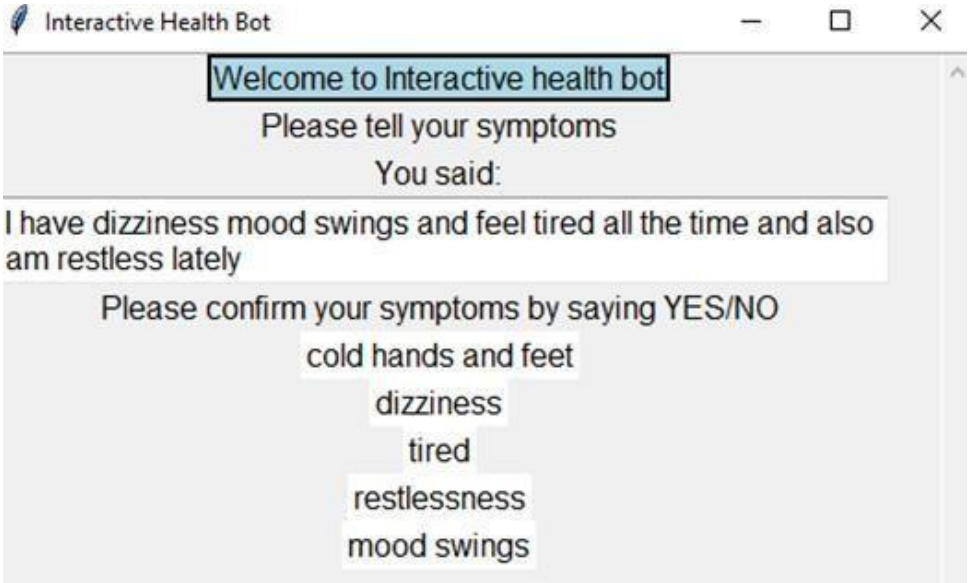

Fig 5.2Verifying the extracted symptoms

The input sentence is processed and the possible symptoms are extracted as shown

in Fig. 5.2. The possible symptoms are displayed and asked for confirmation from the user. This step double checks for only the required symptoms and ignores the not necessary symptoms (Fig. 5.3).

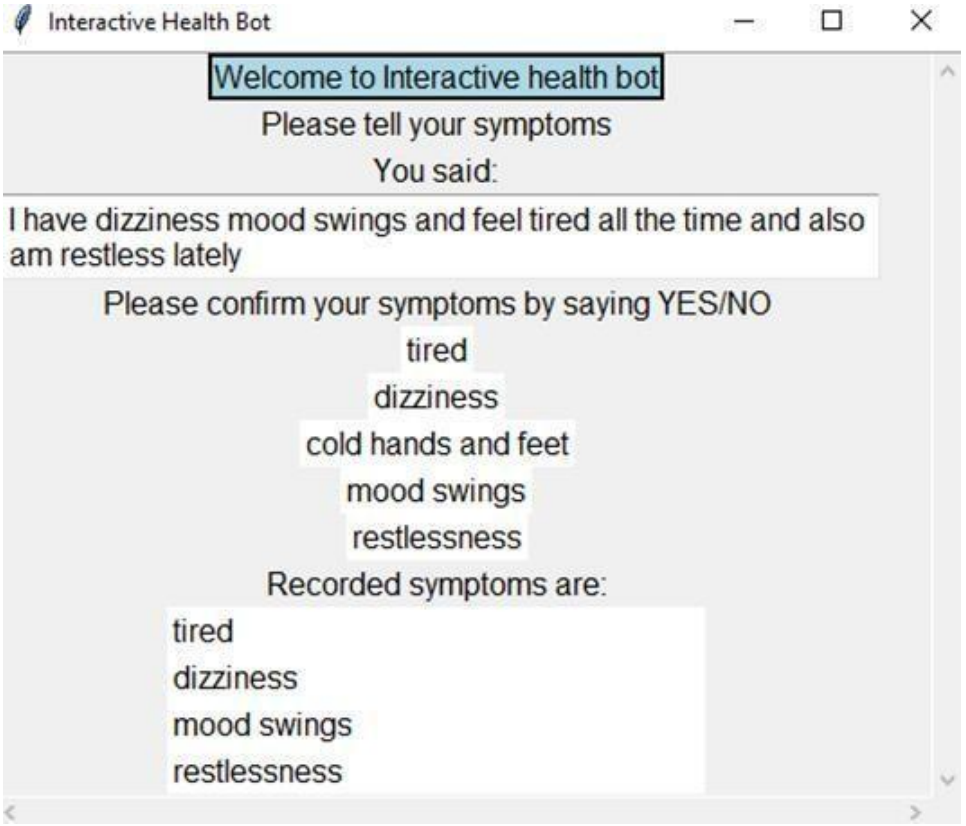

Fig 5.3 Displaying the confirmed symptoms

The confirmed symptoms are displayed. These are the symptoms that will be used further for the classification process (This is out of scope of this article).

Example 2:

Here is an extract of another example (Fig. 5.4) that was considered and the corresponding results obtained.

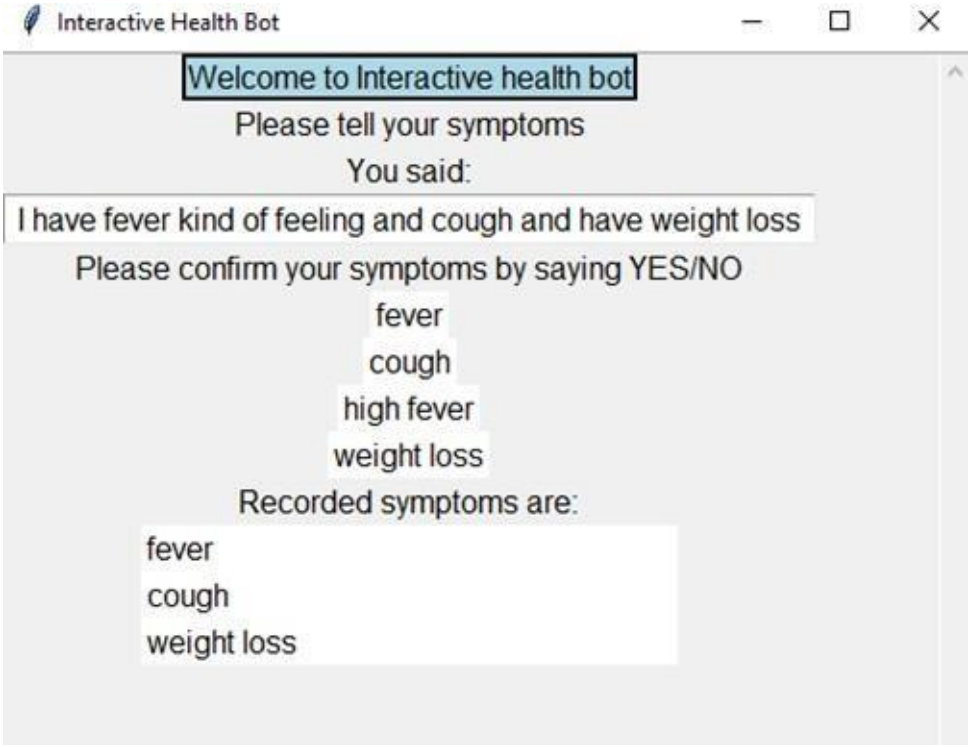

Fig 5.4 Example 2

Upon considering several test cases, the output symptoms were compared with the expected ones and the mean accuracy was found to be 87.53 %. Different variations of the input text were considered and compared with the symptoms that were in our dataset. Fig. 5.5 shows a plot of a few cases along with its accuracy levels.

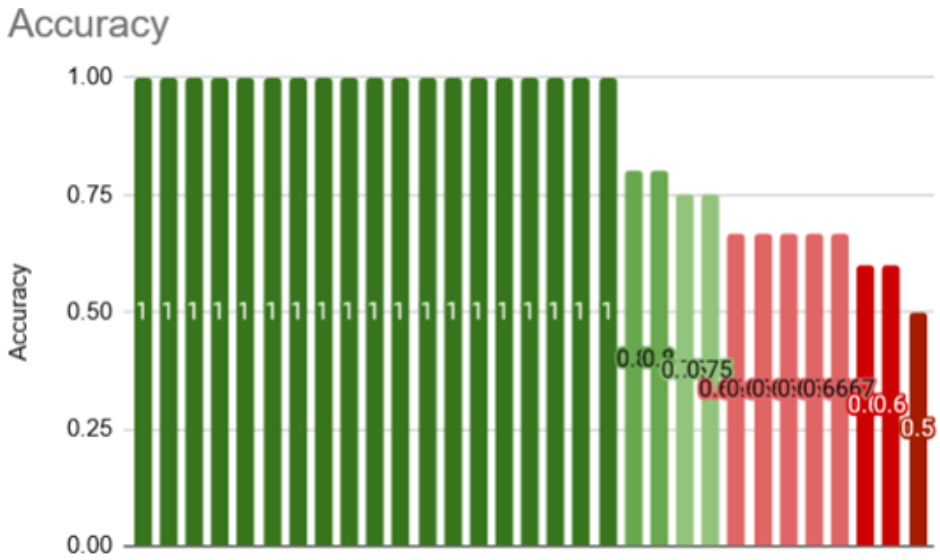

Fig 5.5 Accuracy plot of the proposed system

Similarly, another statistical analysis is done based on the precision and recall which is the F-measure or the F1 score. It is a better measure to analyze the results since

it takes a balance between precision and recall. Hence, it provides a realistic measure of the performance of the system. The Fig.5.6 shows the radar chart of the F-measure. It indicates that most of the values are closer or equal to 1 (which is the ideal value), indicating that the system's performance is good.

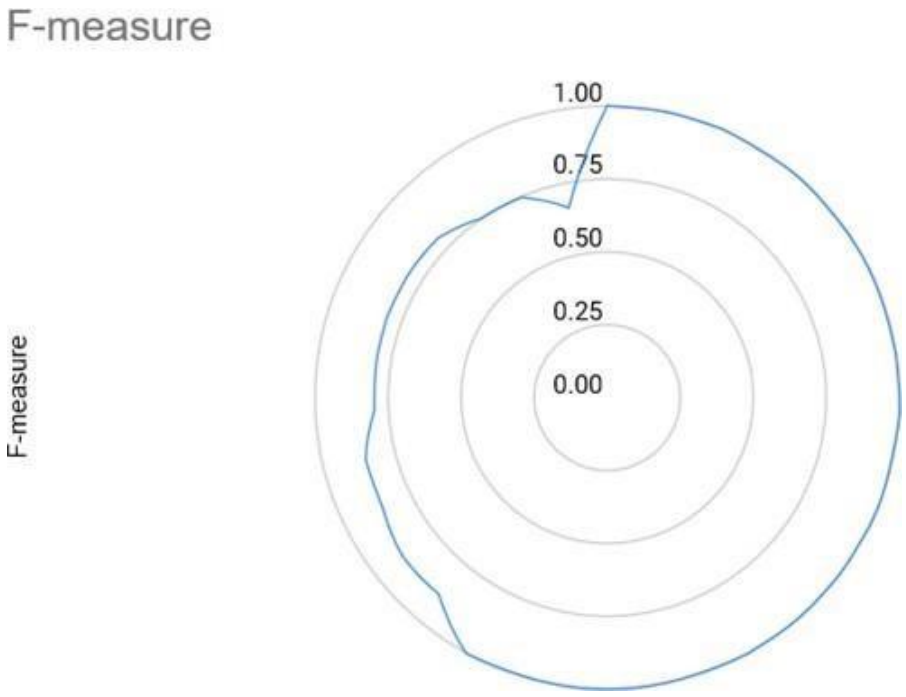

Fig 5.6 F-measure radar plot of the proposed system

5.1.2 CLASSIFICATION

The classification model used is random forest classifier. The classification accuracy of the model implemented was tested using several test cases. All the test case results can be grouped into three different classes. First one being the case when the initial input itself is sufficient to predict the disease. An example of such a case is shown in Figure 5.7.

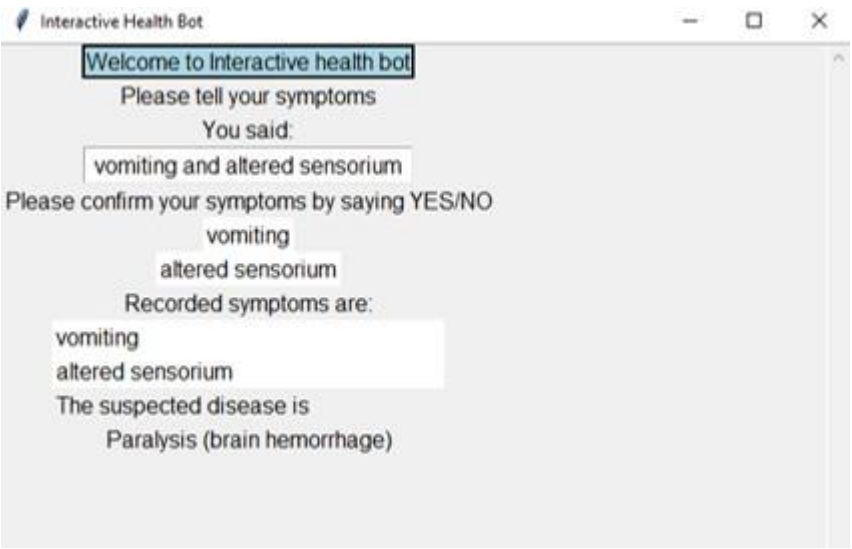

Fig 5.7 Case 1 example

The second case is the one in which it asks further questions about the symptoms the user is facing and eventually predicts the symptom. Figures 5.8,5.9 and 5.10 represent the second case.

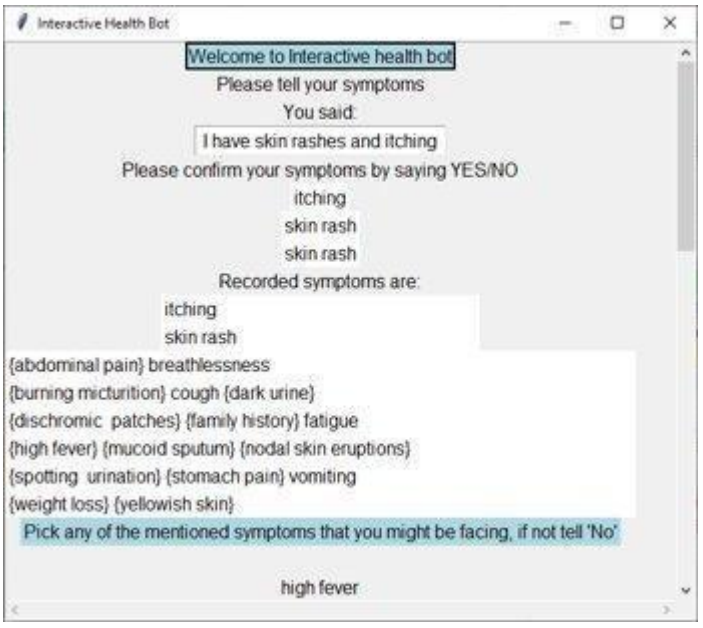

Fig 5.8 Case 2 example input

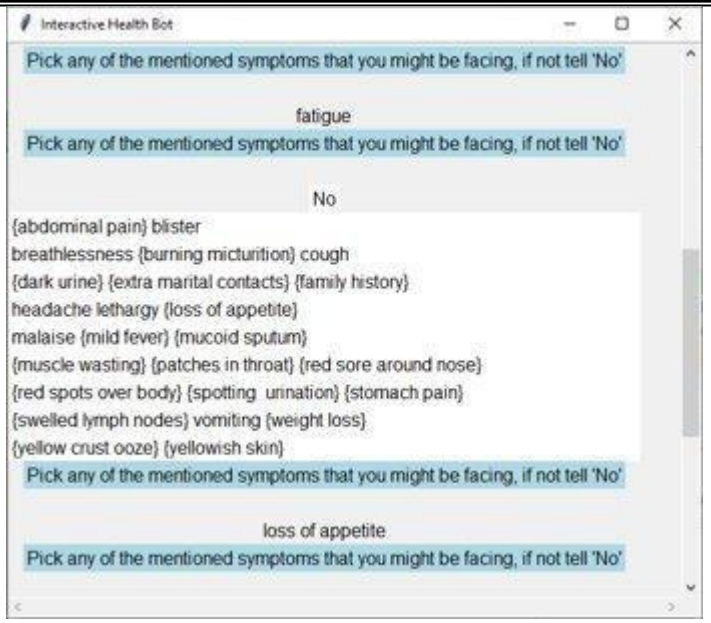

Fig 5.9 Case 2 example questionnaire

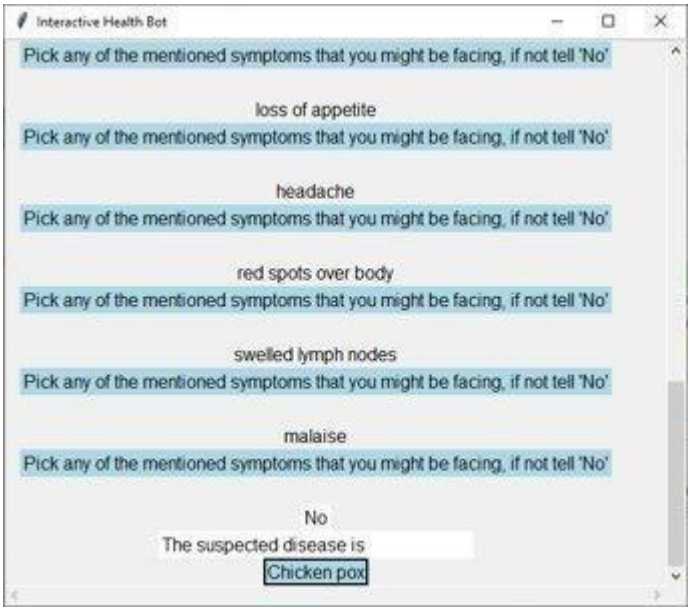

Fig 5.10 Case 2 example result

In the third case, even after asking several questions about the symptoms, the classifier does not predict one possible suspected disease, instead gives a list of probable diseases. Figures 5.11,5.12, 5.13 and 5.14 represent the third case.

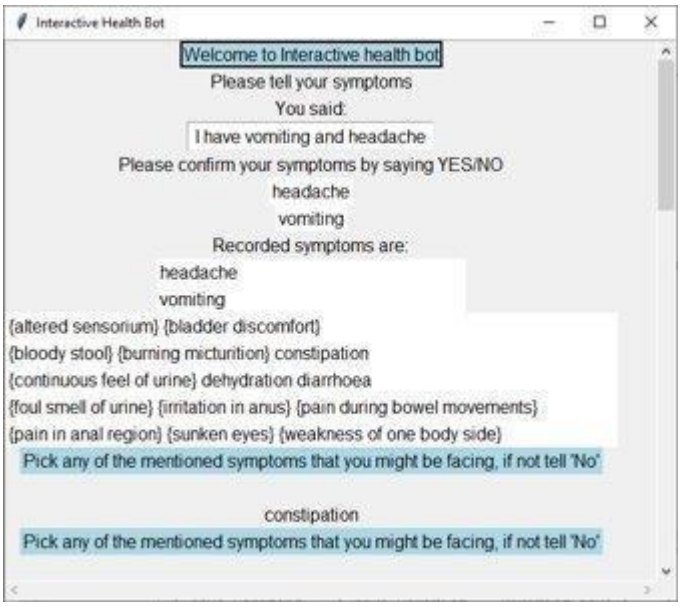

Fig 5.11 Case 3 example input

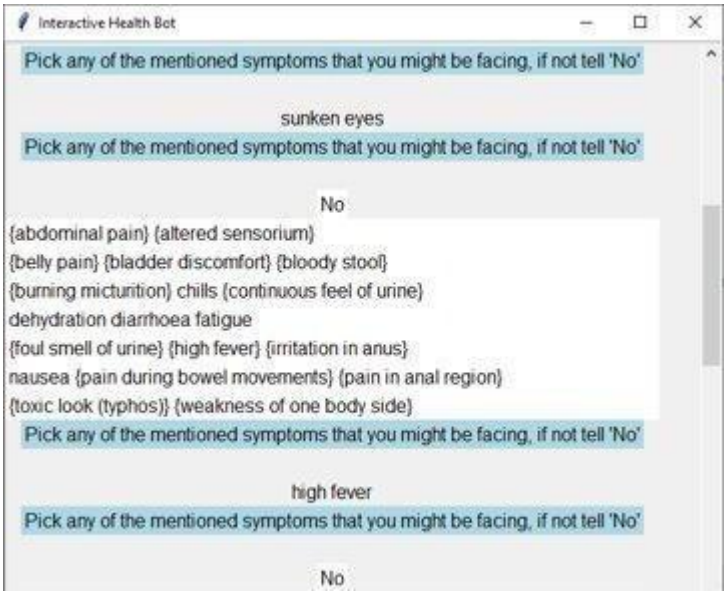

Fig 5.12 Case 3 example questionnaire

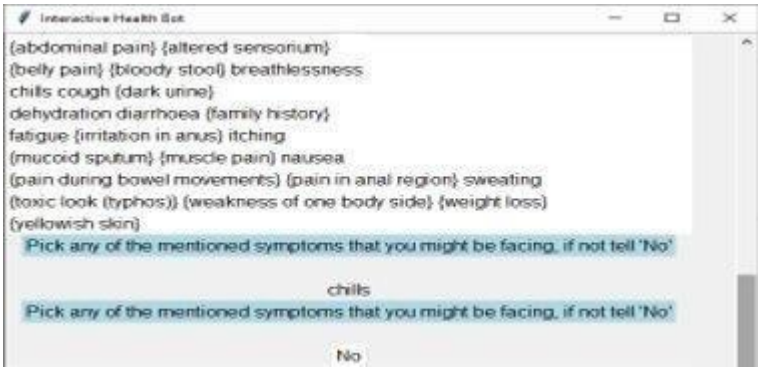

Fig 5.13 Case 3 example questionnaire continued

Fig 5.14 Case 3 example result

These are the three classes of results that were observed. Based on the correct prediction of the disease when legitimate symptoms are provided to the classifier, accuracy of it was measured. Initially while the tests were carried out, several problems were encountered which led to wrong prediction or ambiguous prediction of diseases. Later, these errors were corrected and the same set of inputs was tested again to check if the problems were solved.

In Figure 5.15 there is a stacked bar graph that depicts the number of wrong and right predictions before and after the corrections were made.

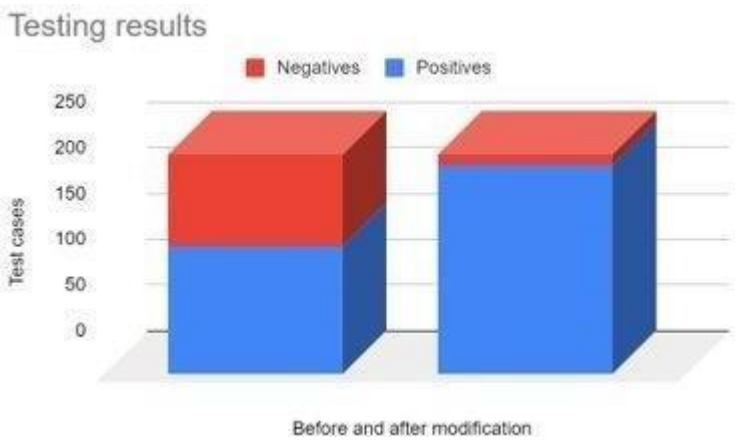

Fig 5.15 Classification testinggraph

Before the modification, there were 140 positives and 101 negative results among the 241 test cases. It accounted to 58.2% accuracy of the classifier. After the corrections were made in both natural language processing part and classification thresholds, the positives went up to 229 with only 12 negatives. The current accuracy of the classifier is at 95.02%.

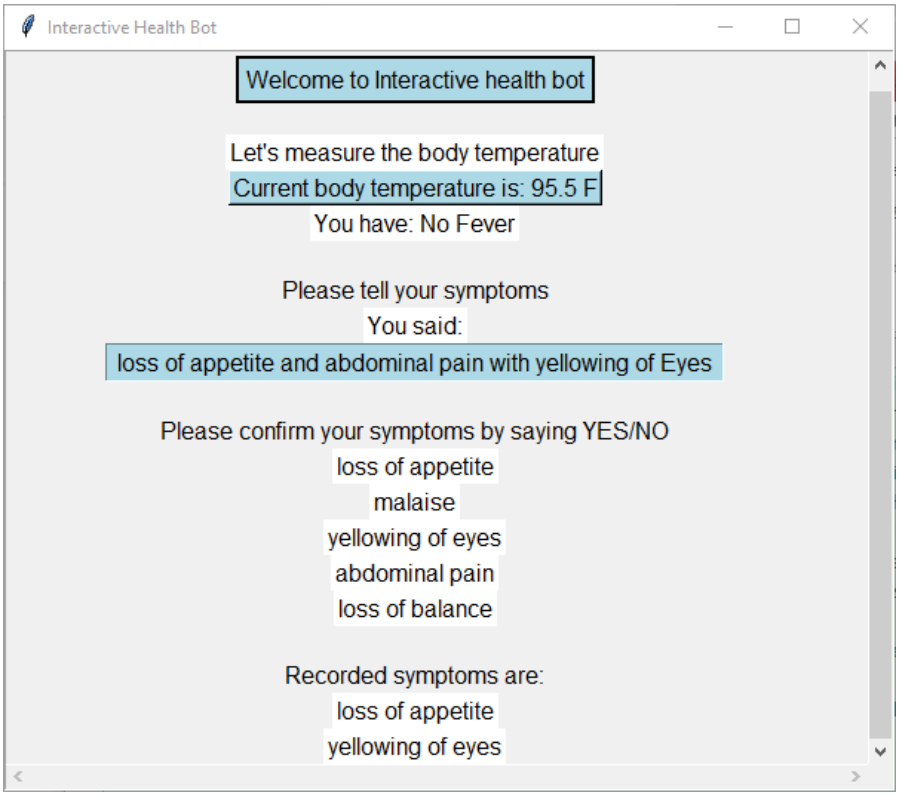

Figure 5.16: Recent GUI window screenshot

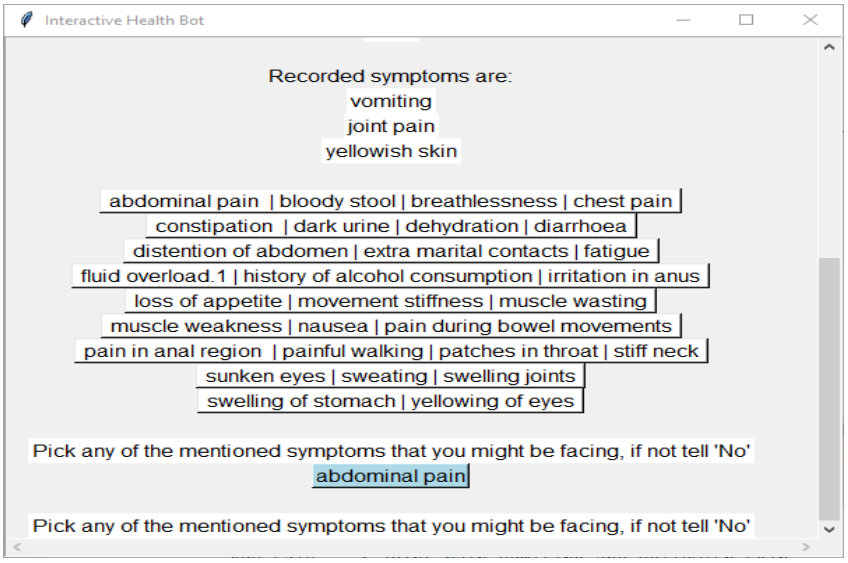

Figure 5.17: Displaying remainingsymptoms

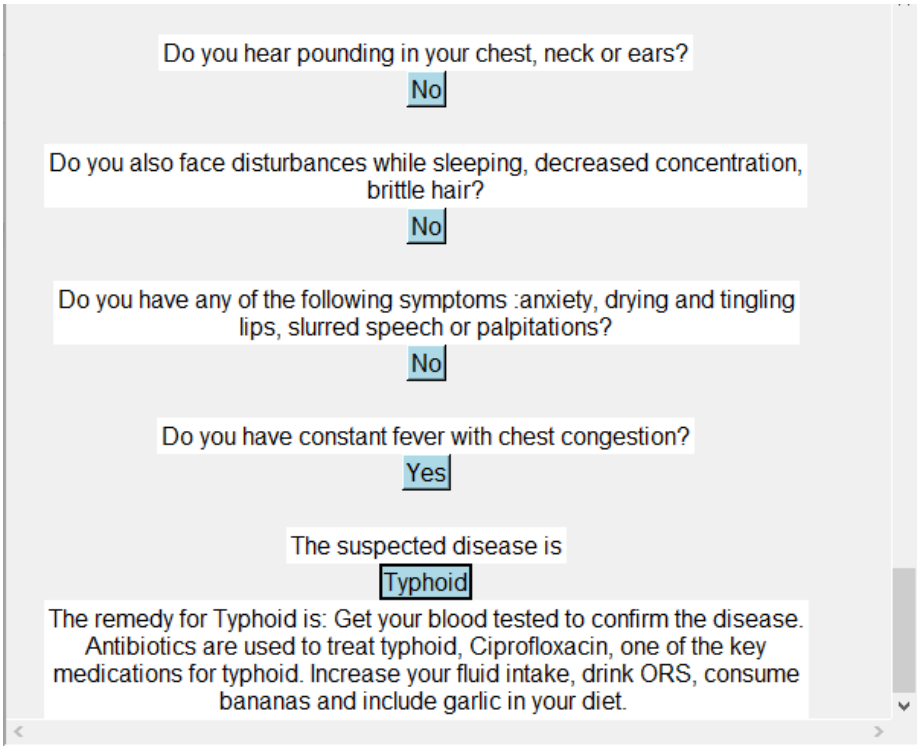

Figure 5.18: Special question implementation and displaying the remedies

In the further modification, we have implemented special questions. We have framed a special question regarding the disease. This is based on research and it aims on dealing with the symptoms that are not there in the considered symptoms. This helps in deciding between two or more similar symptom diseases.

5.2 HARDWARE

5.2.1 DS18B20 TEMPERATURESENSOR(WATERPROOF)

The interactive bot is used for the diagnosis of all sorts of diseases. Most of the diseases consist of fever, be it high or low fever as a symptom. Generally, fever is the precursor to any kind of disease that the humankind can think of. Onset of fever may or may not indicate the presence of a particular disease, but it is very important to detect the presence of fever.

Fever is a common symptom that most of us would have experienced in our lifetime, because of myriad factors. Fever is detected by checking the temperature of a human body mainly with the help of a thermometer. Normal human body temperature is around 98.6^{o} F (37° C). Anything above the normal human body temperature gives rise to feverish condition or simply fever. Fever can be mild or high depending on the body temperature. Body temperature up to 102° C will cause mild fever. Temperature between

102° C and 104° C will lead to high fever. Anything above that, gives rise to high-graded fever which ultimately can lead to the death of the person.

Hence, it is very much necessary to detect the presence of fever beforehand, which can help in the detection of any possible disease quickly thereby saving time and money fora patient.

Our interactive bot comes equipped with a body temperature measurement sensor knownas "**DS18B20WaterproofTemperatureSensor**".Itiscompactindesign, flexible and very easy to use. This sensor can be interfaced to Raspberry Pi very easily where the connection complexity is less. DS18B20 sensor provides the accurate body temperature measurements and few tests were conducted physically to verify the accuracy of the temperature readings. The readings were accurate enough which will help in the detection of fever accurately. Below are the test results of the physical test that was conducted in order to verify the sensor readingaccuracy:

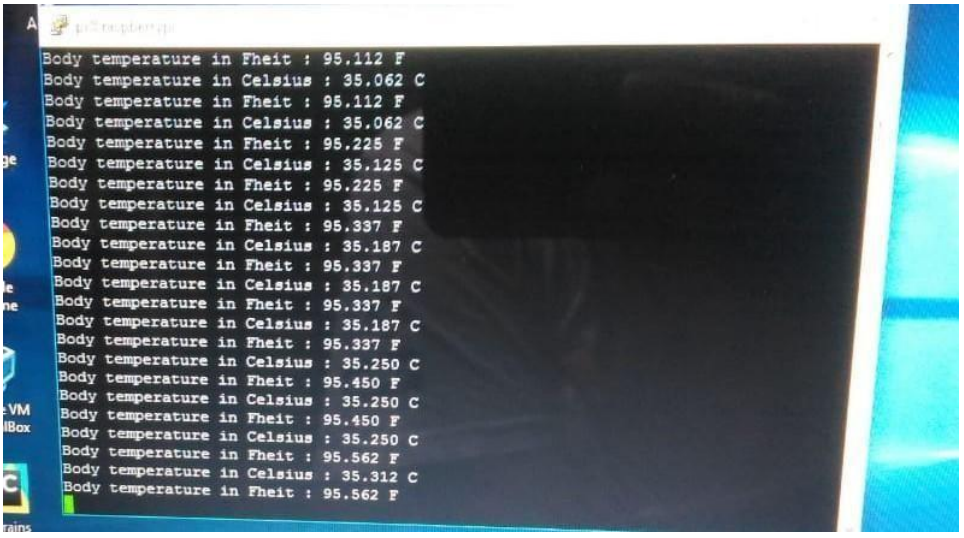

Fig 5.19 DS18B20 Temp. Sensor readings

Asfrom the above testresultsof a –Normaluserwhere the normal body temperature is about 98.6^oF, we can verify the **accuracy and consistency** in the temperature readings which are the factors influencing the selection of a particular device. Since the temperature readings are consistent and accurate enough, the selection of DS1820 Temperature sensor for our interactive bot is justified. These readings will help the user in detection of fever very easily, thus serving its intended purpose. Presence of high or mild fever will speed up the NLP classification process in the detection of a possible diseasequickly.

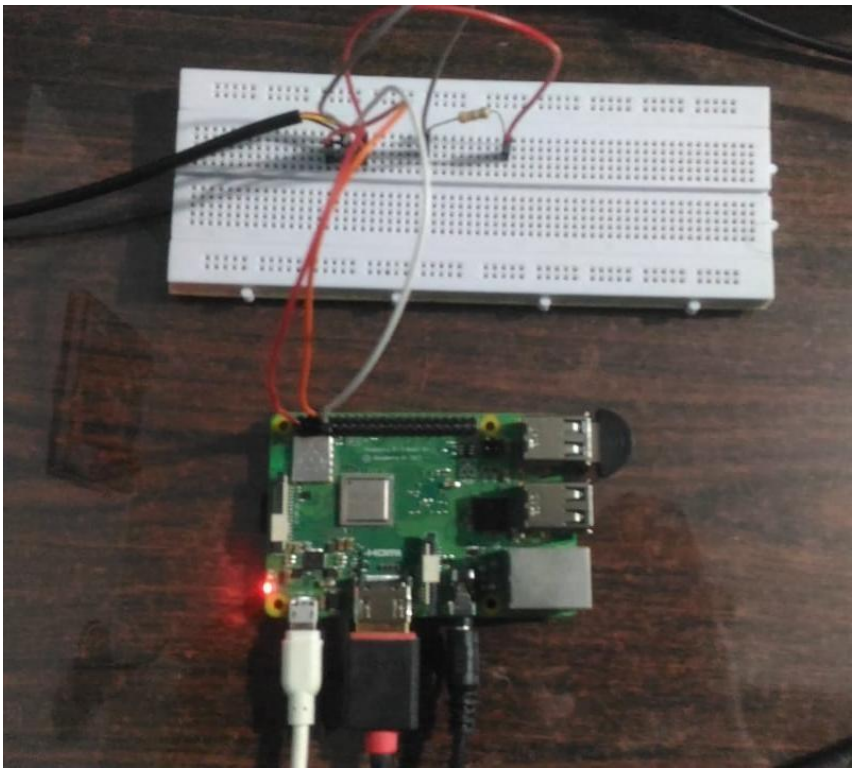

Figure 5.20: Temp sensor circutry

Figure 5.21:USBmic

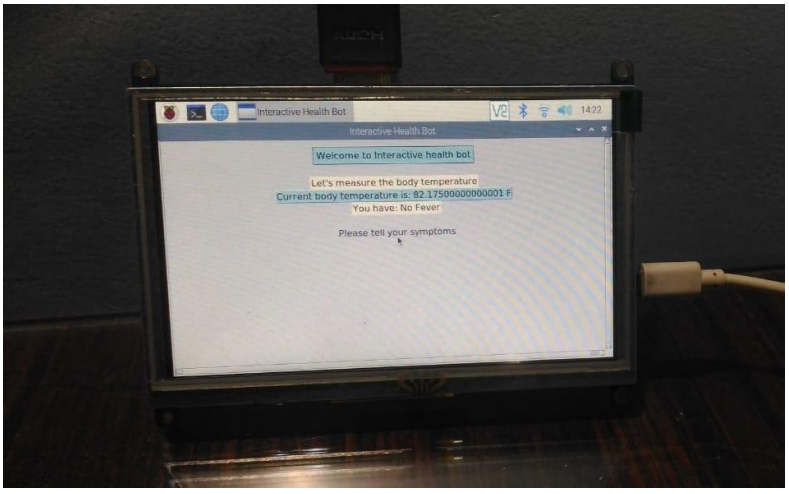

Figure 5.22: LCDdisplay

Figure 5.23: Complete setup

5.3 ANALYSIS

All the entries are recorded in a systematic manner in a Firebase console database. It is as shown below.

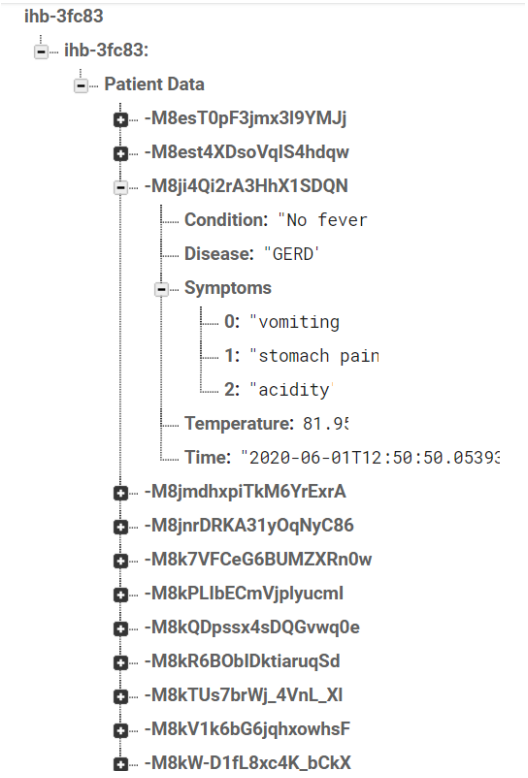

Figure 5.24: Firebase entries

The recorded entities are the diagnosed disease, reading from the temperature sensor, the symptoms entered and the timestamp.

The next figure shows the frequency of the occurrence of each disease in the database entry. This tally gives us a knowledge of what diseases are being recorded the most.

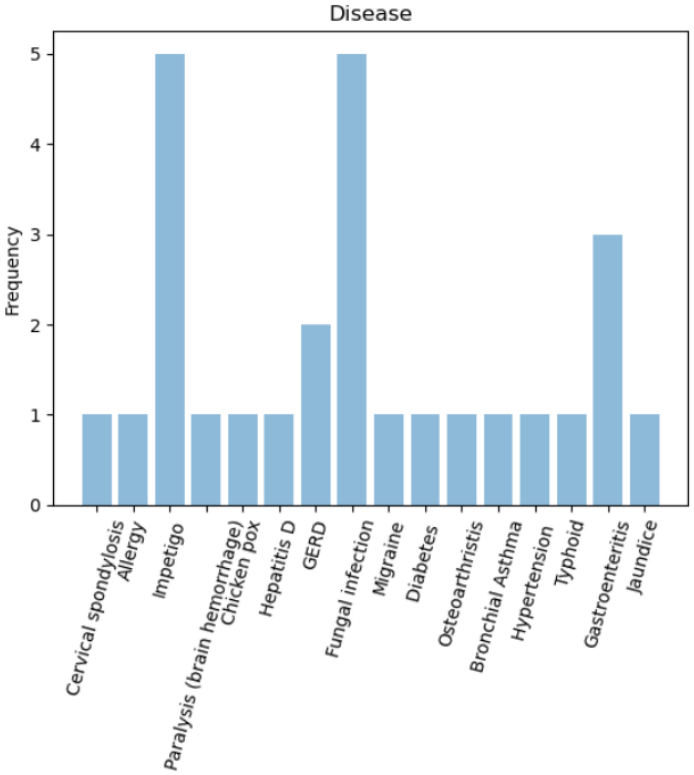

Figure 5.25: Disease tally

Fig.5.26 shows similarly the tally for the symptoms. It gives us an estimate of the most number of symptoms being recorded.

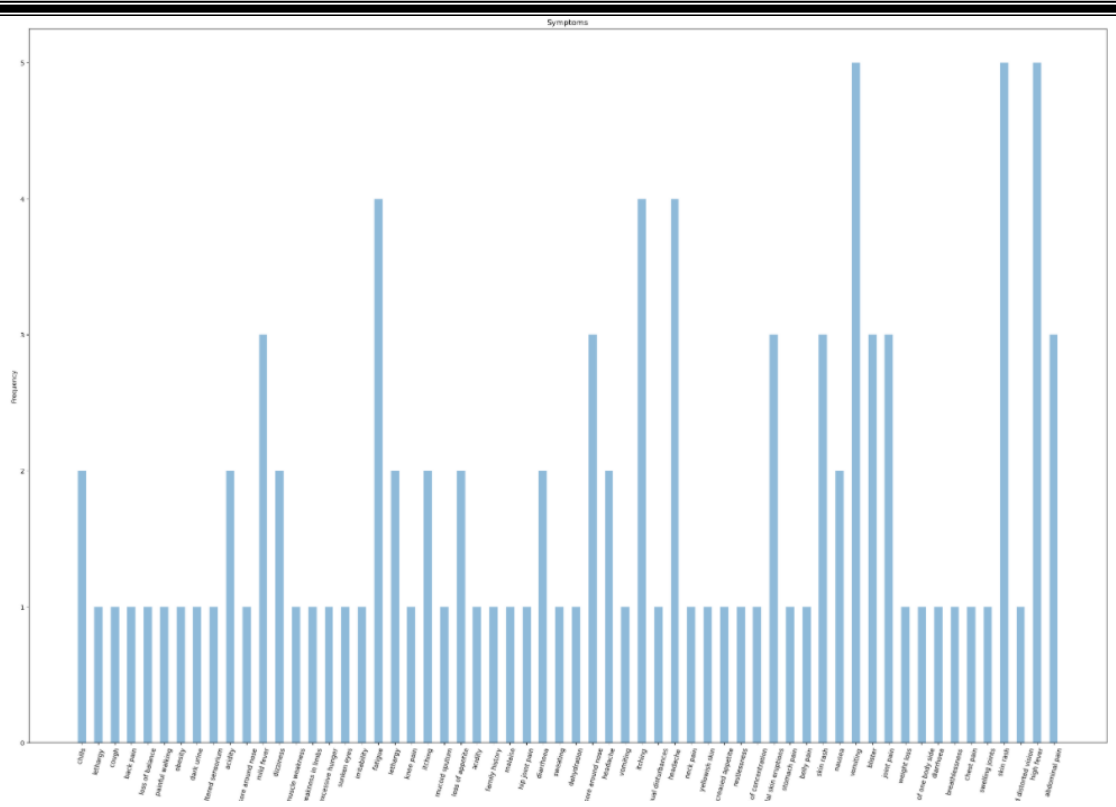

Figure 5.26: Symptom tally

To analyse the frequency of symptoms occurring for each disease, the report of a particular disease taken as an example is shown. This information is pretty useful as it tells us what the most occurring symptoms are for any diagnosed disease.

```
Diabetes
['weight loss', ' fatigue', ' lethargy', ' restlessness', ' obesity', ' increased appetite']
count of symptom in disease:  Diabetes   =  [(' fatigue', 4)]
count of symptom in disease:  Diabetes   =  [(' lethargy', 2)]
count of symptom in disease:  Diabetes   =  [('weight loss', 1)]
count of symptom in disease:  Diabetes   =  [(' obesity', 1)]
count of symptom in disease:  Diabetes   =  [(' increased appetite', 1)]
count of symptom in disease:  Diabetes   =  [(' restlessness', 1)]
```

Figure 5.27: Disease report

CHAPTER 6

CONCLUSION

Chapter-6

CONCLUSION

In conclusion, based on the results obtained, further work will include improving the accuracy of the whole system even with invalid input data by solving all the existing problems. Further work also includes making the user interface more appealing and better to look at. Will be working on converting the software part into a standalone application which can be easily interfaced with RPi. Interfacing the temperature sensor data with the software part. Finally, analyzing the data stored in firebase to gather valuable information.

An interactive bot for predicting diseases using symptoms is proposed and a model which is backed by classification using decision trees and natural language processing is intended to implement. The natural language processing acts on the speech input that is taken and classification acts on the keywords that are output of NLP. Sensors such as temperature and heart rate sensors are interfaced to the raspberry pi which helps in collecting vital parameters. These collected parameters and information are uploaded onto the Google firebase database where analysis on the data is carried on deriving important information.

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APPENDIX

Appendix 1

The test cases for the classification part are given in the link below:

https://docs.google.com/spreadsheets/d/1a6gK3RwuRNkZN0YtkjiFpbghNfz3Eu6s7N-ACz8M64E/edit#gid=0

The proposed method has been tested for a total of 241 cases. The accuracy is calculated for each section, which is used for the overall accuracy. The cases that were positives, i.e. the output was same as the expected, have been coloured green, whereas the other cases that failed to match with the expected output are red.

Appendix 2

The test cases for the NLP part are given in the link below:

https://docs.google.com/spreadsheets/d/13758kAa_BNwbe321NiHPB5TNKdjRJOWaBg6tlE-S3CI/edit?usp=sharing

Appendix 3

We have written two papers one of them is published on IJERT journal in the April issue. The other paper is submitted to the Journal of Critical review. Both the papers are attached to the report.

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Symptoms Extraction from a Voice Input using Natural Language Processing

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***Abstract*— Natural Language Processing is a field that is concerned with interactions between humans and computers using the natural language or the language that humans use. It focuses on making the computers understand the language of humans and to process it to collect information. This field has proved to be very useful in several areas. In this paper we intend to explore the application of natural language processing in the field of primary healthcare. This system is provided with an input regarding the health issues of a person. The system understands the given input and pulls out only the symptoms that are provided by the user which is further utilized for classification purposes. This system was implemented using NLTK Python package and results were displayed using the UI window created using tkinter. The accuracy of this system was found to be 0.8752.**

Keywords*— *Preventive healthcare; Virtual Assistants; Medical information extraction, Natural Language Processing

I. INTRODUCTION

Sickness is not a choice but an uncertain event that happens without knowledge. At times, this illness might be trivial or serious. People, often cannot afford to spend time travelling to a doctor in their busy metropolitan life. To help such people, symptom checkers that provide fast and good aid in improving the health are used [1]. Though there are several symptom checkers yet research in this field is still active as more and more accurate systems are in demand. We set out to develop one such system which is interactive, learn the problems of the user and provides with close to accurate diagnosis along with some remedies.

The vital part of any conversational device is the ability to understand and comprehend what the user is speaking. In other words, defining a set of rules or to be able to teach the device to take in the input, parse it and deduce its meaning. Our project is to build a simple voice-based disease prognosis device (symptom checker). Hence, applying Natural Language Processing (NLP), we create an interface between the user and the core part of the symptom checker (which is the classification of diseases). However, in this paper we focus only on the NLP part of our device.

II. RELATED WORKS

In this section we present literature survey carried out for this work. Literature work from Google scholar and Google search engine were searched with the keywords: preventive healthcare, Virtual Assistants and medical information extraction using NLP. Relevant papers are summarized here.

Emad S. Othman [2], implements a Personal Assistant using Raspberry pi. This Personal Assistant performed numerous tasks like reading out the news, giving information regarding the weather and much more. Primary hardware used is Raspberry Pi to which voice commands are the primary inputs. This device has to be connected to internet continuously, machine learning is also implemented. In the proposed methodology, online modules of speech-to-text and text-to-speech by the AVS are used. The text is query processed online on the wolfram alpha site based on the keywords uttered by the user. Few main keyword modules are wiki, news, joke, weather, etc. This system claims to be customizable as it is using open source modules. Internet is continuously required for performing the tasks. In another work a symptom checker is built using a medical knowledge base and natural language processing. Its medical knowledge base consists of data from open source medical resources, texts, and claims data. This method provides diagnosis using the historical data. Knowledge base is created automatically using structured clinical resources, medical texts, and information extraction engine. Unified Medical Language Systems (UMLS) is used to integrate all the data collected into a knowledge base. The symptom checker includes a front end that collects the user information and a Web server that links all the components present in the checker, it sends the data collected by the front end to the NLP component. NLP component does three tasks, paraphrase generation, negative detection and phrase extraction. A personalize component which provides personalized output based on clinical claims. This symptom checker found discrepancies in classification as KB had more data for one particular class while less for the other. The symptom checker demonstrated an accuracy of 65.9% on an average [2].

Arlene E. Chunget.al.[3], examines the characteristics of health and fitness apps. A number of health-related Voice Activated Assistants (VAA) that are used in a variety of settings for many different functions, such as home monitoring of symptoms or health education. Some studies have suggested using Amazon Alexa to asses deaf speech, provide task support for individuals with cognitive disabilities, and receive voice input from patients to determine "unexpected changes in mood". Such VAA vendor apps are different from the conventional mobile apps, in that the types of interactions are not fully delineated by voice interface because it does not have the same transparency as interacting with a physical user interface. There is also no menu of features or functionalities such that the user could understand the full spectrum of the types of commands one could ask or what kind of information could be provided by the app.

A study was carried out to conduct the descriptive content analysis based on the information provided by the vendors to determine the types of apps in health and fitness categories for

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commercially available VAAs. As per the review, Amazon and Google were the only companies with commercially available hands-free VAAs. The selected apps of hands-free voice activated assistants are screened for health and fitness apps and the filtered ones are taken for review. After evaluation it was found that 309 apps were categorized as "health and fitness apps" by vendors. Their analysis revealed that there were only a limited number of health monitoring apps. This may be, in part, due to the restrictions within Amazon and Google policies for publishing health-related apps. Amazon and Google Home devices have a button that can be pressed to "mute" the device from listening, but instruction manuals do not have explicit language describing that the devices are always listening and that they are recording and storing audio. Also, device and usability should be improved, and focus should be given on interactive user-based interfaces. Apps are released, modified, updated, and discontinued on a regular basis. As a result, there may be apps that were reviewed in this study that are no longer available, and there may have been modifications in app descriptions since data extraction. Despite these limitations, the paper contributes towards an understanding of the characteristics of health and fitness apps available for commercially available, hands-free VAA [3].

Gunjan Dhole et.al.[4], published a paper "Medical information extraction using Natural Language Processing interpretation" which is based on retrieval of medical data from narrative clinical documents using Natural Language Processing (NLP). It explains about how NLP is used to extract medical information from a narrative text by using tools such as tokenizing, noun entity recognizers, parts of speech taggers and relationship extractors. However medical text is different from normal text as it contains complex medical terminologies due to the synonym and disease names. The system that is proposed in this paper tries to understand the text relating to a list of symptoms and it will try to give out the relevant answer.

The authors have clearly stated the need for using NLP in this case. According to them most information retrieval systems are rule based systems as against NLP which can extract data from free text form. Several medical extraction systems have been researched and mentioned about in the paper.

Information Retrieval:

NLP stores and interprets meaning at both query and the document. A general text retrieval system would have the following flow:

1. Document processing
2. Query processing

a) Query matching
b) Ranking and sorting

The knowledge base stores all the medical documents. The system consists of two modules, document and query processing. The answer processing module will check which type of query is coming in. If its "Definitive" it will follow the flow for definitive queries. If it's "Descriptive" query, it will follow the flow for descriptive queries. Document Processing and retrieval is done from the Query processing module which gives noun entities as output.

Lots of research is ongoing in the field of extraction of medical data using NLP. It needs different tools as compared to the normal ones. The system described in this paper does not store the information or the medical history anywhere and the extraction techniques can be improved further [4].

Gustav Cederblad [5], in "Finding synonyms in medical texts – Creating a system for automatic synonym extraction from medical texts" focuses on bridging the gap between laypeople and medical professionals by extracting the synonyms from a corpus consisting of medical texts and to evaluate whether these words are actually related. Because, sometimes the more specific medical words are described by a more commonly used familiar word. The aim is to find synonyms and related words for different diseases. In the unigrams that are considered after tokenizing only the noun entities are retained since synonyms belong to the same part of speech (noun).

They give a stark description about the semantic relationships and weighting, which deals with giving importance to words occurring in the same context, close to the target word, giving a hint about the word's true meaning. Work that has been already done in synonym extraction has been reported.

The project is implemented in stages. At the first stage unwanted data is got rid of, in the medical corpus, leaving only the necessary information. Misspelled or unknown characters are taken as noise which should be reduced, and hence regular expressions are used which filters out irrelevant data. The second stage is a part of speech filtering. They make use of a SNOMED dictionary describing diseases, words are chosen from that, which are all nouns. They only focus on unigrams. Hence, filtering out all the non-noun (NN) words would lead us to the synonyms that are actually required. The pos tagger – NN is made use of. However, it is not 100% accurate.

Since, in large pieces of text, it is very likely that the same word will appear in different forms, NLP techniques like stemming and lemmatization are utilized. Distinguishing between words having the same root word is unwanted since it is only the meaning that is of interest to the analysis. Since Stagger has already been used for the part-of-speech tagging, lemmatizing the corpus is not more computationally costly than stemming. That is because all the lemmas have already been extracted and it is just to replace its original word in the corpus.

Later, they go about extracting the semantically related words getting a csv file as an output consisting of the most similar words for each SNOMED term. The analysis is further done. The use of a questionnaire is stated to interact with the user to narrow down to disease they are conveying. A human evaluation on a few selected members is performed and the results and the statistics are explained in the paper.

The limitation of this system was that it only handles unigrams which could decrease the overall performance of the system. Bigrams or further ngrams would also describe the adjectives for a particular symptom, like "runny nose" or "severe pain". This emphasizes the need for a model that can manage n-grams larger than unigrams. The noise removal is also not very efficient in retaining only the noun parts of the user input. When replacing synonyms in texts, it is imperative

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that no information gets lost, especially in a medical context where correct information could be the difference between life and death. Dealing with this sensitive and personal information is difficult and entails many ethical questions that have to be considered. Hence there is much room for improvement on the aforementioned system [5].

III. PROPOSED METHOD

For a device to accept voice input and understand it, certain steps have to be followed to parse the input. The proposed method includes the following steps:

1. Speech synthesis
2. Modifying the dataset
3. Interpreting the input
4. Review

We have made use of NLTK package to perform the tasks, due to its easy to use interfaces such as WordNet, and a suite of text processing libraries for tokenization, stemming, tagging, parsing and so on, all of which shall be explained below. It is a free open-source package. Pandas library is a fast, flexible and a powerful tool that can be used for reading datasets in our project.

The block diagram of the proposed method is as shown in Fig.1.

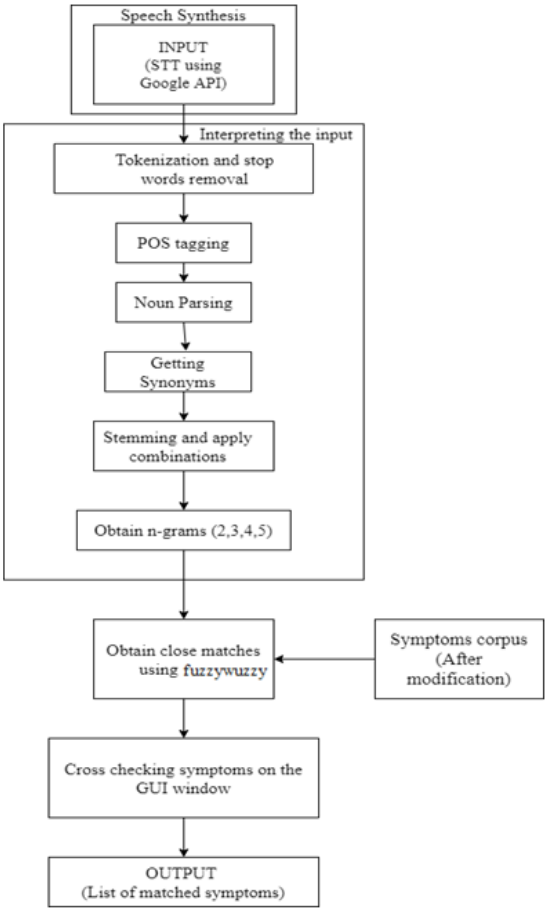

Fig.1: Block diagram of the proposed system

A. *SPEECH SYNTHESIS:*

Starting off with the process, the prime component is the speech. It is random and cannot be directly used. Hence, an analog to digital conversion is done to digitize it to convert it from the raw physical sound to an electrical form with the help of a microphone. Once this is done, there are several models that transcribe the audio into the textual form. There are several packages/ APIs to perform this.

We have made use of one of the popular ones, that is, Google Text to Speech API (gTTs). It is a convenient tool. The Speech Recognition library retrieves the audio input and is very flexible. The Recognizer class of Speech Recognition recognizes the speech and has many methods that it makes use of, using different APIs, like gTTs. It requires an internet connection but is quite facile.

We would also need an additional tool for the reverse, that is text to speech. The pyttsx module does text to speech conversion and also works offline.

B. *MODIFYING THE DATASET:*

Any symptom checker would have a dataset of a list of diseases and symptoms as a basis for prognosis. We start out by considering a small dataset, which could be a .csv file and is modified slightly. Since the user input will not be in the exact terms as in the symptom's dataset, we take the root words of the symptoms. Hence, if the user's words have the same root word(s) of that of the dataset, it matches. Taking the root word is done by processes like stemming or lemmatization. They are collectively called as text normalization processes of NLP.

Example: The words 'suffering', 'suffered', 'suffers' have the same root word i.e., 'suffer'.

Stemming deals with reducing the word to its root, even if the root isn't a meaningful word. 'Porter Stemmer' of the 'nltk.stem' package is one of the oldest and popular one that is in use. It is known for its simplicity and speed.

Lemmatization, however, reduces the word to a meaningful root word which is called as lemma. Python NLTK provides 'WordNet Lemmatizer' that uses 'WordNet Database' to lookup lemmas of words. This modified dataset will be kept as a reference for further steps.

C. *INTERPRETING THE INPUT:*

Here comes the interesting part! Making sense out of the input and obtain the relevant information (i.e. symptoms) that can be made use for classification. The raw input is a string from the speech synthesis. All the actions are performed on this string.

One of the important steps of any NLP method is text tokenization. It involves splitting words or sentences from a body of text into entities. It is carried out by 'word_tokenize' module.

Example: Input – "I am facing severe pain in the chest"
Output – ["I" , " am" , "facing" , "severe" , "pain", "in" , "the" , "chest"]

Stop words are those words that are insignificant and are usually filtered out because they contain a vast amount of unnecessary information. The common ones used in English language are 'as', 'that', 'of' and so on. The 'stopwords' corpus of nltk gives the collection of such words. Hence, they are removed from the tokenized list.

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Next, we do what is called as part of speech (pos) tagging. It is also called as grammatical tagging. The goal of a pos tagger is to apply a linguistic tag or information to the tokens. Since we're interested in only extracting the symptom part of the text, which are commonly forms of nouns or adverbs (mostly), we filter in only such tagged words. Our process becomes more efficient step by step.

Some common WordNet methods are applied like that of 'lexeme', which gives the tense forms of a particular word.

Another important factor to be considered is a lexical relation called synonyms. Since the limited dataset obviously cannot cover all the similar meaning words that it contains, we need to simply find the same for the input and compare it with the dataset. The 'synset' package provides a list of synonym words.

Example: Input - "ache"

Output - {'ache', 'aching', 'hurt', 'languish', 'pine', 'smart', 'suffer', 'yearn', 'yen'}

We will also be considering two, three, four or n-tokens together and apply all of the above-mentioned methods, because the symptoms could also be a group of words. They are called as n-grams and n could be any positive integer. The commonly used as bi and tri-grams. They are compared against the dataset.

To be on the safer side, we will use a comparing algorithm on the input n-grams with the dataset symptoms, to get the closest matching words. Such an algorithm is made use in the 'fuzzywuzzy' module of Python. It is based on the Levenshtein distance to calculate the difference between the given two texts or strings. It has a value that is used as a threshold to compare and provides all the closely related words, completing our text parsing.

Several other methods can also be applied to make the output more accurate; however, the efficiency and the time taken must be kept in mind. The output from each and every stage of all the NLP processes is compared with the dataset symptom and any closed match is recorded in an output list.

D. REVIEW:

Since we always must err on the right side, the recorded output list of symptoms is cross-checked with the user. A yes/no question-answer type interface is made for each symptom and recorded again in a list. If the output is an empty list, the user is asked if they would want to start over or exit. Finally, the verified output list is the result of the NLP performed on the input and is passed on to further classification processes.

Additionally, a general user interface (GUI) is built with the help of 'tkinter' Python library. The NLP will be running in the backend while a GUI is present in the frontend providing the user a visual experience along with audio updates.

IV. RESULTS

The proposed methodology was implemented using Python (version 3.7.3) and user interface was developed using tkinter package and the results obtained are presented in this section.

The confirmed symptoms are displayed. These are the symptoms that will be used further for the classification process (This is out of scope of this article).

Example 2:

Example 1:

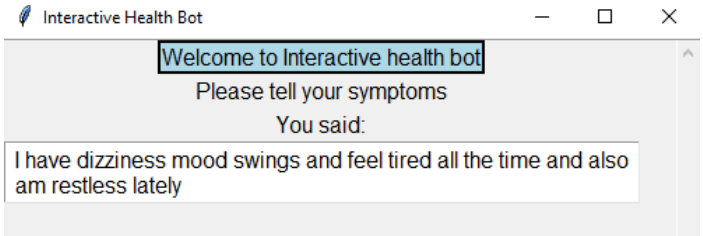

Fig.2: Recording the user input

The voice input is taken into the system. Fig. 2 shows an example of an input sentence describing the symptoms of the user.

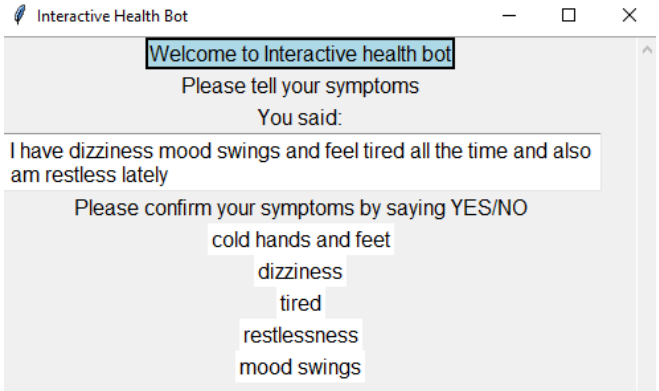

Fig.3: Verifying the extracted symptoms

The input sentence is processed and the possible symptoms are extracted as shown in Fig. 3. The possible symptoms are displayed and asked for confirmation from the user. This step double checks for only the required symptoms and ignores the not necessary symptoms (Fig. 4).

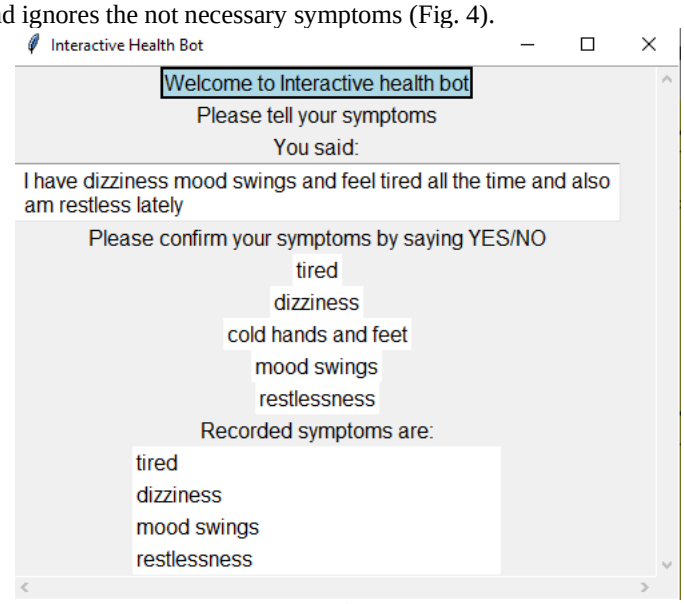

Fig.4: Displaying the confirmed symptoms

Here is an extract of another example (Fig. 5) that was considered and the corresponding results obtained.

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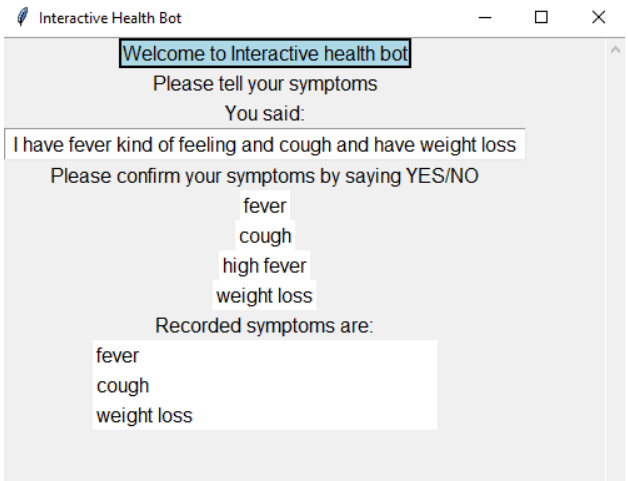

Fig.5: Example 2

Upon considering several test cases, the output symptoms were compared with the expected ones and the mean accuracy was found to be 87.53 %. Different variations of the input text were considered and compared with the symptoms that were in our dataset. Fig. 6 shows a plot of a few cases along with its accuracy levels.

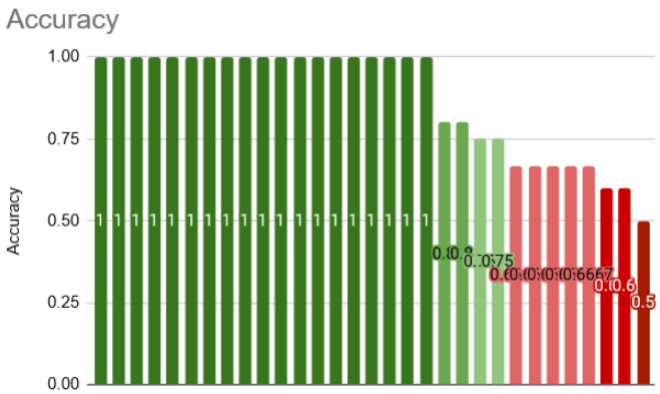

Fig.6: Accuracy plot of the proposed system

Similarly, another statistical analysis is done based on the precision and recall which is the F-measure or the F1 score. It is a better measure to analyze the results since it takes a balance between precision and recall. Hence, it provides a realistic measure of the performance of the system. The Fig. 7 shows the radar chart of the F-measure. It indicates that most of the values are closer or equal to 1 (which is the ideal value), indicating that the system's performance is good.

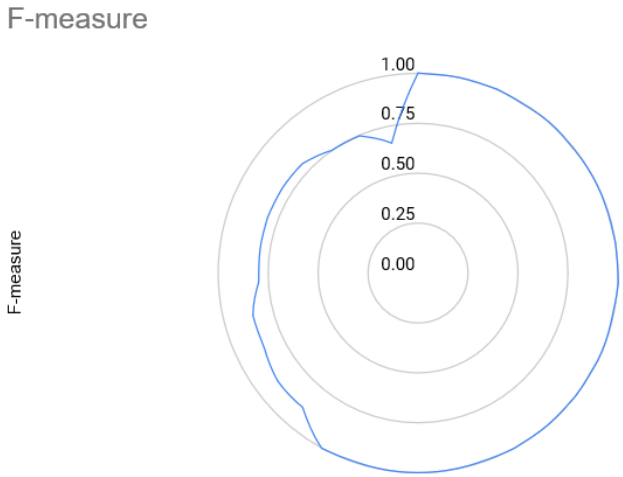

Fig.7: F-measure radar plot of the proposed system

V. CONCLUSION

A system for extracting relevant symptoms from the user voice input was put forth. The primary use of the extracted symptoms is to help in classifying what medical condition the user is facing. The system proposed makes use of many natural language Python packages and implementation steps to parse the input. The testing was done for several cases and the accuracy turned out to be good. Since healthcare is one of the growing industries in the world, our system proves to be a useful method that can be applied to provide a good prognosis prediction. Future work could include various ways to refining the recorded symptoms to improve the accuracy and user dependency.

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Voice Controlled Personal Health Assistance Device using Natural Language Processing and Random Forest Disease Classifier

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Abstract

Health assistant devices are a need of the hour. In this paper, we outline one such device that has been developed. It is voice controlled device that uses Google's speech to text converter API and Natural Language Processing for the communication with the user. The user provided input symptomsare classified into a disease using a Random Forest classifier. The NLP and classification are performed within the Raspberry Pi which acts as a standalone device. Temperature sensor is also interfaced for better user experience. The developed classifier has the accuracy of 95.02% which has further scope of improvement with a larger data set.

Keywords: *Health Assistant, Random Forest classifier, Disease classification, Firebase, Natural Language Processing, DS18B20 Temperature Sensor*

1. Introduction

The access to medical care is unacceptably low worldwide in spite of the increasing demand for medical care due to population aging and increasing burden of diseases. Especially in resource limited places, patients do not have access to medical care in a timely manner. Thus, tools to help patients with self-diagnosis and self-triage is urgently needed to mitigate the shortage of medical care resources. Health monitoring devices are the need of the hour in this era where the number and kinds of diseases are on rise. All the diseases have to be treated as soon as possible and a remedy has to be provided to stop it from making more severe damage.

Conventionally, when people fall sick, they try to be the doctor themselves and ignore their issue by swallowing a few tablets to get rid of it for the time being. But these issues might pile up later causing them bigger trouble. To help people get sooner diagnostics and help them take more care about their health, this study has come up with an idea of a health-aid device. The study attempts to effectively provide a potential diagnosis for patients and direct them to the appropriate care setting.

The device serves to facilitate self-diagnosis and to assist with triage. The diagnosis deals with typically educating patients on the range of diagnoses that might fit their symptoms. The triage informs patients whether they should seek care at all and, if so, where and with what urgency (that is, emergently or within a few days).

For patients with a non-emergent problem that does not require a medical visit, our device can reassure and recommend they stay home and prevent them from using unnecessary antibiotics. Reducing the number of visits saves patients' time and money, deters overprescribing of antibiotics. Logging of all the diagnosis and the symptoms that led to it and other relevant information would help in statistical analyzing and interpretation.

We propose to implement a solution to help the user with primary care medical advice connecting symptoms with ailments with complete storing of the diagnosed information in a cloud-based database for any further analytics. Basically, it makes a diagnosis of a

disease based on the data on the prevalence of disease and its sensitivity and specificity of symptoms. In the future, such devices would help keeping up the quality of primary care. Remainder of the paper is organized as follows: Section 2 presents the Literature Survey, Section 3 explains Materials and Methods, Results and Discussion in Section 4 and conclusion in Section 5.

2. Literature Survey

Several articles have been published with different methods to assist patients. This sectionpresents the literature that has been relevant to this study.

A hybrid sensing and wearable device for emergent medical situations with individual modules monitors blood glucose, blood-pressure, heart rate, body temperature, and ECG [1]. It is capable of taking necessary initiatives for emergency medication and treatment in case of unpredictable situations and notifying the patient about routine medication like a medical assistant according to physician prescription. Contribution towards an understanding of the characteristics of health and fitness apps for commercially available hands-free voice activated assistants (VAA) is also available [2].

For such applications, Natural Language Processing (NLP) has been used particularly the extraction of symptoms from the input sentence. Emad S. Othmanhas proposed a Raspberry Pi based personalized voice command system which basically takes the audio input of the sound using a microphone and is further passed through the system to search for keywords. The modules work on the essence of searching for keywords and giving output by matching keywords. Once the symptoms put forward by the user have been processed, the corresponding text output is converted to speech using the text to speech converter and the audio output is sent to the user using the speaker [3]. Having researched about the different methods available to aid the patients, some specific domains tend to serve better in building a unique solution to the problem. The paper published by researchers Gunjan Dhole and NileshUke [7], is based on retrieval of medical data from narrative clinical documents using NLP. Since medical text is different from normal text as it contains complex medical terminologies due to the synonym and disease names, the system proposed in this paper tries to understand the text relating to a list of symptoms and gives the relevant answer. The system described in this paper does not store the information or the medical history anywhere and the extraction techniques can be improved further. Gustav Cederblad [8] focuses on bridging the gap between laymen and medical professionals by extracting the synonyms from a corpus consisting of medical texts and to evaluate whether these words are actually related. That is because sometimes the more specific medical words are described by a more commonly used familiar word. It aimed to find synonyms and related words for different diseases. However, it handles only unigrams, which could decrease the overall performance of the system. Hence, it must be improved to manage n-grams larger than unigrams. The initial symptoms extraction process from the audio voice input is as detailed. The whole process is divided into four steps namely, Speech synthesis, modifying the dataset, interpreting the input and review [11].

Next thing is about classifiers and [9] discusses on what and why decision trees are used in Data Mining and some of the most popular and widely used algorithms used to create decision trees, namely: ID3 (Iterative Dichotomiser), C4.5, CART and Random Forest. Based on the advantages and drawbacks of each of the methods, Random Forest Classifier was shown to be a better match for this study.

From several studies in [6], we can deduce that the Raspberry Pi is an inexpensive, fully customizable, and programmable credit card-sized single board computer, which supports a large number of peripherals and network communication. Therefore, it is suitable for small scale prototype testing and is very efficient and cost inexpensive.

Healthcare database differs from a traditional database. Health care database includes a lot of information and fields that have to be created and managed [10]. M. Sathya, S. Madhan, K. Jayanthi in their paper on IoT [4] focus on addressing the lack in database connectivity between the different cloud environment in monitoring the data in constant time intervals and to analyze data. A cloud based IoT system is proposed that can be implemented in different health monitoring systems. In the paper [5], a Raspberry Pi based IoT model is presented which uses docker container instead of virtual machine for the server. Pi accepts data from sensors, and these are transferred to the cloud using server and users can access the database from the cloud. Since it uses a docker container, it intends only in sending the data to different users and nothing about the analysis of the data.

Having reviewed the literature and analyzing the advantages and disadvantages discussed in the above research papers, the proposed method absorbs the necessary and useful elements from each one of them and it also works on the preferable performance improvements.

3. Materials and Methods

This section presents the proposed work in the form of block diagram, NLP, classification and hardware details.

3.1. Block diagram

The proposed technique is conceptualized as shown in Figure 1. The complete system is embedded on a Raspberry Pi 3 Model B+ (RPi) board, which acts like a standalone device performing all the necessary computations.

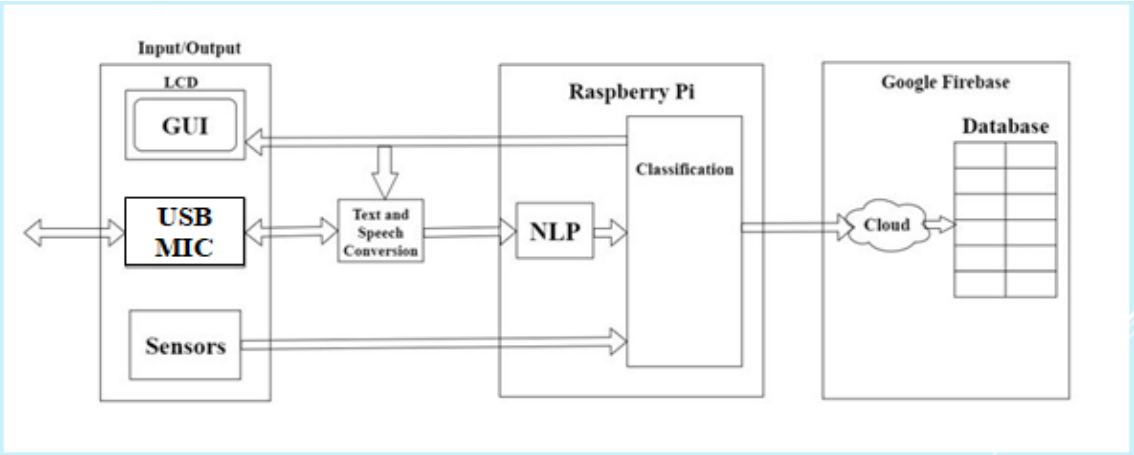

Figure 1.Proposed Block Diagram

The proposed device is interfaced with sensors. The device takes voice commands and processes those commands to provide details about a disease when the symptoms are provided to the device. It poses the users a series of questions about their symptoms or requires users to input details about their symptoms themselves. The symptoms are classified into a disease using a decision tree and an audio and text output is provided to the end user at each and every step until it drills down to the closest disease that is matched. The output includes the disease related information, and any of the common treatment procedures if it is not a serious one. The temperature sensor acts as one of the

symptoms measuring interfaces that are present on the device. The implementation includes NLP, classification and hardware details.

3.2. Natural Language Processing (NLP)

The basic idea for the input part is to implement a voice system that can take in the symptoms the user is facing and to incorporate it into the classification part to output the estimated prognosis clearly. The user interacts with the device via speech input which will be captured by an USB microphone mounted on the RPi. Since the RPi Model 3B+ doesn't inherently possess a microphone for capturing audio input, we use an USB microphone for capturing the speech input. The audio is converted into the text format using an API, which is then broken down into segments to understand and analyze. Hence, text normalization is done accordingly with the aid of appropriate natural language processing tools like stemming, synonym extraction, tagging, and so forth. Nonetheless, there could be some symptoms that the speech-to-text API has failed to recognize properly or even, some fallacies in the extraction process leading to any of the symptoms not being caught from the user input. On that account, we make use of some string-matching algorithms to perfect extraction of the symptoms from the input.

We make use of two phonetic algorithms, i.e. Soundex and Metaphone. Soundex is an algorithm that depicts how two words are related depending on their English pronunciation. It converts the input string into a 4 character long alphanumeric value which is compared with the Soundex value computed for the other string. The key values are the same for two similar sounding words. Whereas Metaphone, which is a refinement on the Soundex algorithm, has a wider set of English pronunciation rules allowing for various lengths of keys. For example: The Soundex and Metaphone values of the following two symptoms are:

A = 'muscle wasting'	B = 'muscle paining'
Soundex(A) = 'M242'	Soundex(B) = 'M241'
Metaphone(A) = 'MSKL WSTNK'	Metaphone(B) = 'MSKL PNNK'

From the above example, when compared on the basis of how similar sounding two words are, we note that the Soundex values differ by 1 whereas the second part of the Metaphone values of both the words are not very similar, indicating that Metaphone gives more accurate results. Hence, using a combination of these algorithms we capture the symptoms that would otherwise have gone unrecognized.

Several other methods can also be applied to make the output more accurate; however, the efficiency and the time taken must be kept in mind. The output from each and every stage of the NLP process is compared with the dataset of symptoms that is stored a priori in the RPi and the matched symptoms are recorded in an output list. This list is further presented to the classifier.

3.3. Classification

To begin with, a model based on simple binary Decision Tree was built. It was a model with a node matrix for the training and testing of the Data Frame. Each symptom present against its disease was marked as "1" else as "0". Although the model was built successfully, there were issues faced during entering of the symptoms as well as the predictions. Due to the above stated problems, the model was dropped.

The second model that we tried was based on Random Forest Classifications. It used Label Encoder for pre-processing and fitting the transform of the target variable to achieve classification. It took in symptoms one at a time as either "1" or "0" as Yes or No respectively. Although the model was promising in results, the process of entering inputs

was tedious and long. Further, it also was not compatible with the input methods used by the NLP section of our project and our proposed hardware. Thus, this version of the classifier was also dropped.

Due to the promising results shown by the second model, it was decided for us to continue using Random Forest Classifier as our classifier. Elements of the previous model such as Label Encoder for pre-processing, fit transform of the target variable and consideration of target variable probability was kept intact. To achieve a model based on our requirements and proposed idea many changes were made.

Bagging (Bootstrap Aggregation) -Decisions trees tend to be sensitive to the data they are trained on, small changes to the training set can result in significantly different trees or tree structures. Random forest takes advantage of this by allowing individual trees to randomly sample from the dataset with replacement, resulting in different trees. This process is known as bagging.

Random forest takes advantage of this process of randomization of data to each tree, which leads to higher tolerance to variance overall. It means that the variance levels may be varied for different trees but that doesn't act as a tradeoff when taking the forest in its entirety.

3.3.1. Working of the Random Forest Classifier:

Before any classification process, the data has to be prepared and pre-processed as standard procedure.

This is achieved by pre-processing package imported from package sklearn (Python). The method known as Label Encoder from the pre-processing package is used for data prepping. The Label Encoder is used to label encode and convert the data in machine readable form. It assigns a unique number (starting from 0) to each class of data. Here, the encoded data is the target variable or the "Prognosis" and used in fitting and transform fitting the model. For the Random Forest implementation, the Random Forest Classifier package is imported from sklearn. This model makes use of transform fitting by the package to fit the data. The transformed data being the target variable, "Prognosis".

The number of trees is an important factor in forest building. From testing, it was observed that number of trees i.e. n_estimators = 10 gives the better results (as opposed to the other number of trees we tested). Thus, the number of trees in our model is 10 and is given as input attribute to the Random Forest Classifier function used.

From the package sklearn, metrics and classification_report is also imported. In the model, four methods of the package Random Forest Classifier is used namely: fit, fit_transform, predict_proba and classes.

3.3.2. Steps in classification:

Using the symptoms given by the user initially, the code creates a list of the symptoms covered and a new data frame. The symptoms list created appends each symptom one at a time after checking if such a symptom exists in our training set of symptoms. The new data frame created such that it contains only columns of the entered symptoms.

1. Further, using the above data frame, probability of the possible diseases is determined. It checks each disease that has one or more symptoms of the entered symptoms and predicts the probability of the new list or class of diseases.
2. On calculation, a new list of the possible diseases is created. As an additional step to narrow down on the prediction, the disease with least or minimum probability is removed. At this stage, to get the disease names in user readable form inverse transform is taken to cancel the transform done at the beginning of the prediction.

3. On the output side, the symptoms covered, the possible diseases, probability of possible diseases and the remaining symptoms are displayed as lists.
4. The probabilities of the diseases are checked, if one of them is above 0.75 then the suspected disease is returned and breaks from the loop.
5. Else, if none of the probabilities are above 0.75, there exists a loop to allow the user to add more symptoms. This step is prompted by remaining symptoms of the possible diseases in the list.
6. There is also a provision to not add any symptoms by giving input as “None”.
7. This loop continues for 3 iterations. If within 3 rounds, a disease has probability above 0.75, it is returned as suspected disease and loop is broken.
8. Else if by the end of 3 iterations, a singular suspected disease is not narrowed down and if none of the possible diseases have probability above 0.75, then the loop is broken.
9. In such a case where disease has not been able to be classified even after all the iterations, there exists a special round with our training data specific questions to ultimately point to a particular disease.
10. At the end of classification, the disease name and its remedies are displayed on the output screen.

3.4. Hardware

The block diagram is as shown in Figure 1. The proposed work is implemented using Raspberry Pi 3 Model B+ (RPi) board, which acts as the heart of the device. Starting off with user interface part, the user interacts with the device via speech input which will be captured by a USB microphone that is mounted on the RPi. Since the RPi in general doesn’t inherently possess a microphone for capturing audio input, we use USB microphone for capturing the speech input. The audio is converted into the text format using speech to text convertor and is fed to the NLP software. Here the text is parsed, and the required keywords are extracted and matched with the dataset of disease-symptoms that is stored a priori in the RPi. The matched keywords are then passed to the classification algorithm that takes place using a tree which filters out the closely matching disease through further classification as per the user inputs. The classification and the predicted diseases are all displayed on a 5 inches HDMI LCD screen through Graphical User Interface (GUI) and also audio output facility is available on connecting a speaker or a headphone or an earphone to the RPi through the 3.5mm jack present on the RPi. Additionally, sensor DS18B20 Waterproof Temperature sensor is interfaced with the RPi in order to capture vital signs like temperature, which are transferred into the classification block. Having done the prediction of the disease along with the remedial information alongside, all the interaction data that includes the prognosis, the symptoms that led to the prediction and other information are all pushed to Google firebase.

DS18B20 Waterproof temperature sensor:DS1820 waterproof temperature sensor serves to retrieve body temperature data by placing the tip of the sensor on the armpit. The result of body temperature detection is received by Raspberry Pi and processed through conversion from temperature to voltage. The resulting voltage changes will adjust to the condition of the body temperature of the patient at that time. The result of a translation made by Raspberry Pi will be displayed on the HDMI display attached to the Raspberry Pi. All patient data stored on Raspberry Pi can be called or viewed on the portable or personal computer.

The DS18B20 sensor interface with the Raspberry Pi is simple as shown in the figure below, making it a highly portable device. Connection complexity is less and the sensor is easy to handle, thereby ensuring optimum comfort when it is used for measuring the body temperature.

DS18B20 temperature sensor testing is done by placing the end of the sensor or pairing the sensor on the armpit for one or two minutes. Armpit is a body part that can use as a basis in the sensor readings. The readings of the body temperature sensor on the HDMI display in the form of Fahrenheit reading indicate that the sensor works well. Furthermore, to obtain accuracy testing of temperature sensor data, a digital thermometer was used as a comparison in getting the error rate. The success of this test can be seen on the HDMI display or by portable readings. The measurement results displayed on the HDMI display is an indicator that the sensor DS18B20 works appropriately.

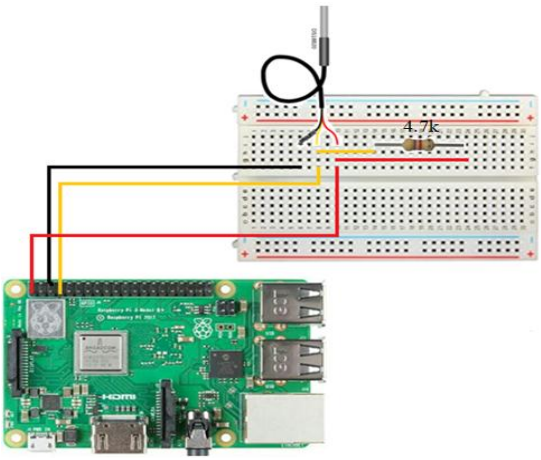

Figure 2. DS18B20 Temperature Sensor Circuitry

4. Results and Discussion

The proposed methodology was implemented using Python (version 3.7.3) and the graphical user interface (GUI) was developed using tkinter package and the results obtained are presented in this section. The voice input is taken into the system. The input sentence is processed and the possible symptoms are extracted. The possible symptoms are displayed and asked for confirmation from the user. This step double checks for only the required symptoms and ignores the not necessary symptoms. The confirmed symptoms are displayed. These are the symptoms that will be used further for the classification process.

The classification model used here is random forest classifier. The classification accuracy of the model implemented was tested using several test cases. All the test case results can be grouped into three different classes. First one being the case when the initial input itself is sufficient to predict the disease. The second case is the one in which it asks further questions about the symptoms the user is facing and eventually predicts the symptom. In the third case, even after asking several questions about the symptoms, the classifier does not predict one possible suspected disease, instead gives a list of probable diseases.

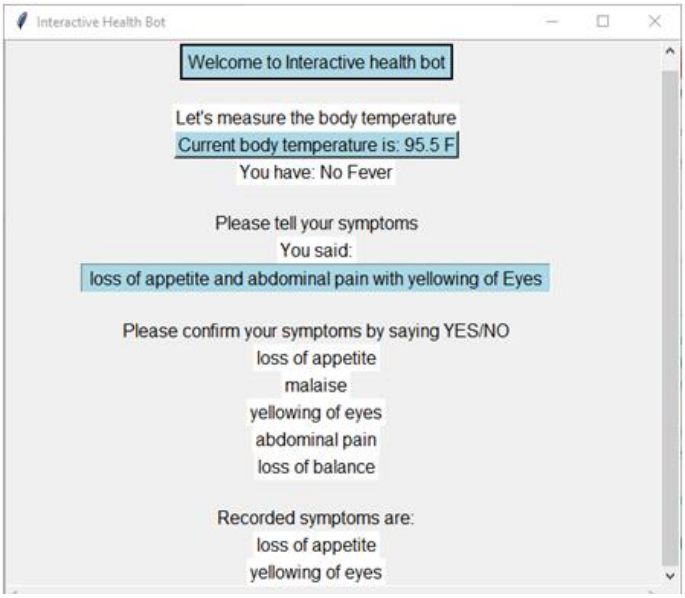

Figure 3. GUI Window Screenshot

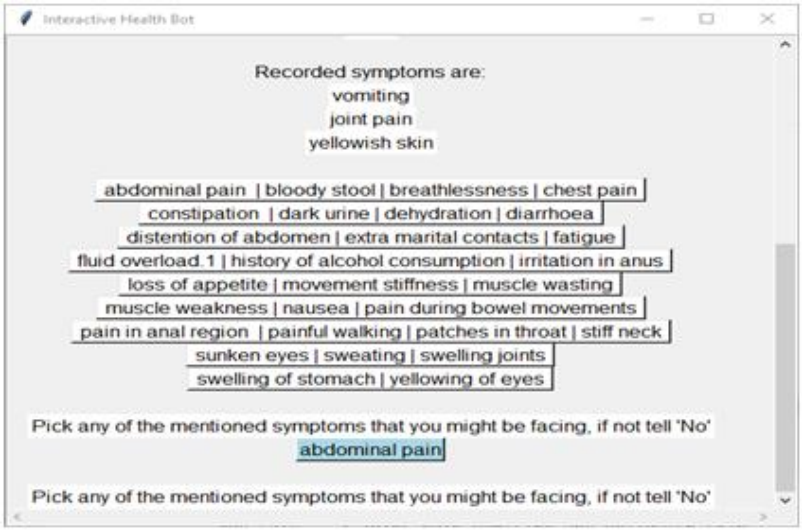

Figure 4. Displaying Remaining Symptoms

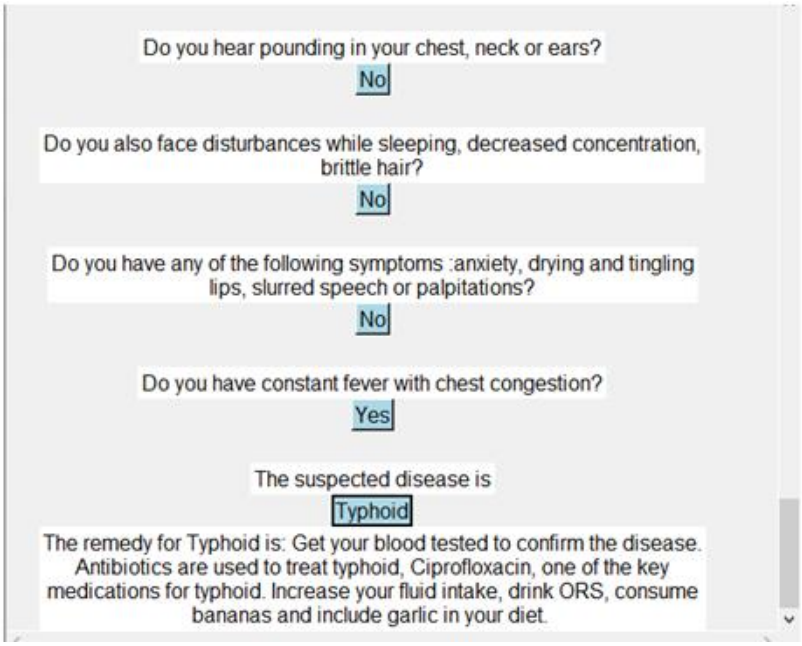

Figure 5. Special Question Implementation and Displaying the Remedies

In the further modification, we have implemented special questions. Special questions were framed regarding the disease. This is based on research and it aims on dealing with the symptoms that are not there in the considered symptoms. This helps in deciding between two or more similar symptom diseases.

For classification, based on the correct prediction of the disease when legitimate symptoms are provided to the classifier, accuracy of it was measured. Initially while the tests were carried out, several problems were encountered which led to wrong prediction or ambiguous prediction of diseases. Later, these errors were corrected and the same set of inputs was tested again to check if the problems were solved. In Figure 6, there is a stacked bar graph that depicts the number of wrong and right predictions before and after the corrections was made.

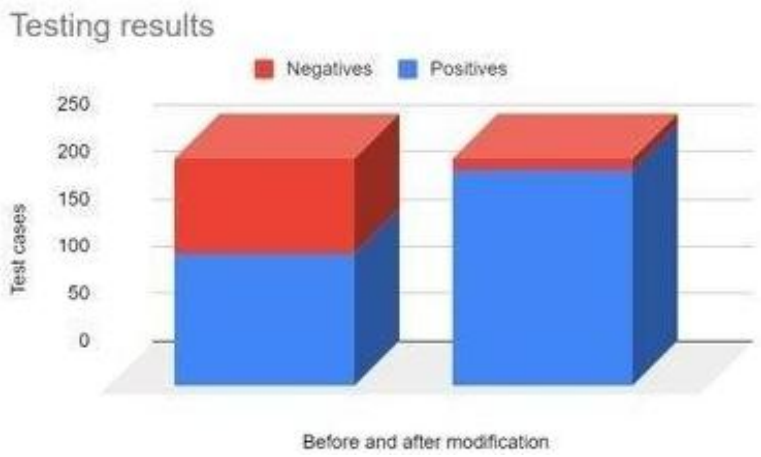

Figure 6. Testing Results

Before the modification, there were 140 positives and 101 negative results among the 241 test cases. It accounted to 58.2% accuracy of the classifier. After the corrections were made in both natural language processing part and classification thresholds, the positives went up to 229 with only 12 negatives. The current accuracy of the classifier is at 95.02%.

All the entries are recorded in a systematic manner in a Firebase console database. It is as shown below. The recorded entities are the diagnosed disease, reading from the temperature sensor, the symptoms entered and the timestamp. The next figure shows the frequency of the occurrence of each disease in the database entry. This tally gives us knowledge of what diseases are being recorded the most.

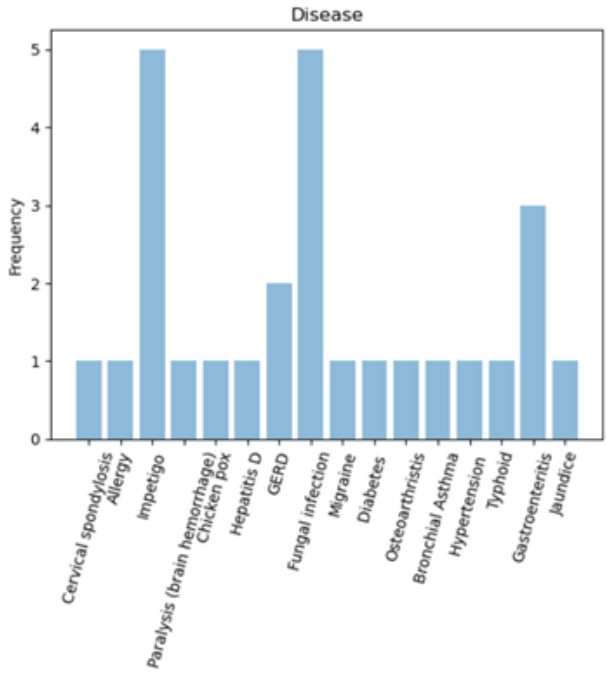

Figure 7. Disease Tally

Figure 8 shows similarly the tally for the symptoms. It gives us an estimate of the most number of symptoms being recorded.

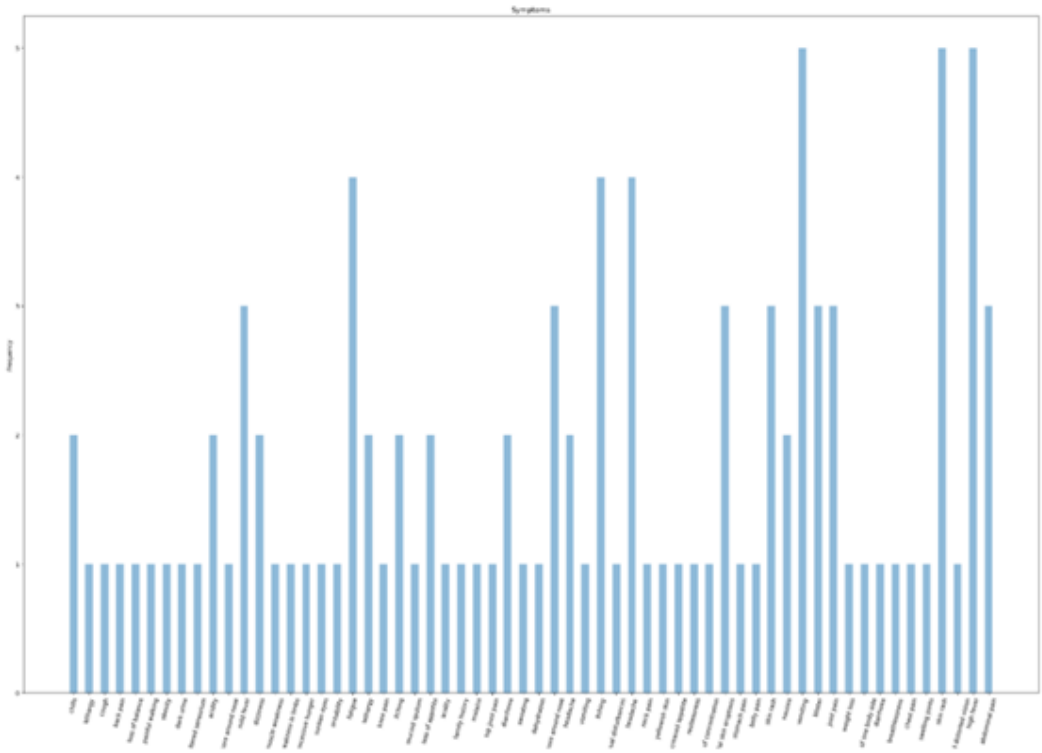

Figure 8. Symptom Tally

To analyze the frequency of symptoms occurring for each disease, the report of a particular disease taken as an example is shown. This information is pretty useful as it tells us what the most occurring symptoms are for any diagnosed disease.

```
Diabetes
['weight loss', ' fatigue', ' lethargy', ' restlessness', ' obesity', ' increased appetite']
count of symptom in disease:  Diabetes   =  [(' fatigue', 4)]
count of symptom in disease:  Diabetes   =  [(' lethargy', 2)]
count of symptom in disease:  Diabetes   =  [('weight loss', 1)]
count of symptom in disease:  Diabetes   =  [(' obesity', 1)]
count of symptom in disease:  Diabetes   =  [(' increased appetite', 1)]
count of symptom in disease:  Diabetes   =  [(' restlessness', 1)]
```

Figure 9. Disease Report

The interactive bot is used for the diagnosis of all sorts of diseases. Most of the diseases consist of fever, be it high or low fever as a symptom. Generally, fever is the precursor to any kind of disease that the humankind can think of. Onset of fever may or may not indicate the presence of a particular disease, but it is very important to detect the presence of fever. Fever is a common symptom that most of us would have experienced in our lifetime, because of myriad factors. Fever is detected by checking the temperature of a human body mainly with the help of a thermometer. Normal human body temperature is around 98.6^o F (37° C). Anything above the normal human body temperature gives rise to feverish condition or simply fever. Fever can be mild or high depending on the body temperature. Body temperature up to 102° C will cause mild fever. Temperature between 102° C and 104° C will lead to high fever. Anything above that, gives rise to high-graded fever which ultimately can lead to the death of the person. Hence, it is very much necessary to detect the presence of fever beforehand, which can help in the detection of any possible disease quickly thereby saving time and money for a patient.

Our interactive bot comes equipped with a body temperature measurement sensor known as “DS18B20 Waterproof Temperature Sensor”. It is compact in design, flexible and very easy to use. This sensor can be interfaced to Raspberry Pi very easily where the connection complexity is less. DS18B20 sensor provides the accurate body temperature measurements and few tests were conducted physically to verify the accuracy of the temperature readings. The readings were accurate enough which will help in the detection of fever accurately. Below are the test results of the physical test that was conducted in order to verify the sensor reading accuracy:

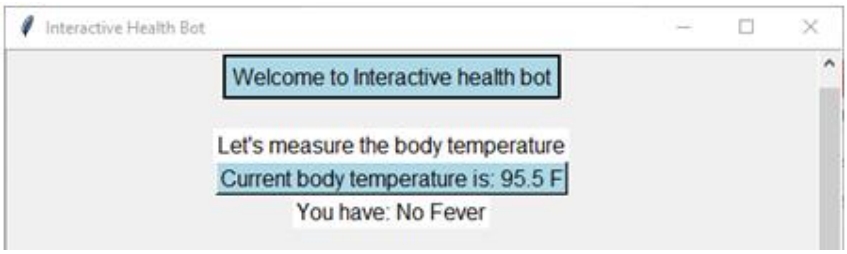

Figure 10. DS18B20 Temperature Sensor Readings

As from the above test results of a normal user where the normal body temperature is about 95.5^o F, we can verify the accuracy and consistency in the temperature readings which are the factors influencing the selection of a particular device. Since the temperature readings are consistent and accurate enough, the selection of DS1820 Temperature sensor for our interactive bot is justified. These readings will help the user in detection of fever very easily, thus serving its intended purpose. Presence of high or mild fever will speed up the NLP classification process in the detection of a possible disease.

5. Conclusion

An interactive bot for predicting diseases using symptoms is proposed and implemented. The model uses natural language processing and decision trees for classification. The natural language processing acts on the speech input that is taken and classification acts on the keywords that are output of NLP. A temperature sensor is interfaced to the Raspberry Pi which helps in collecting a vital parameter. The collected parameter and all the related information regarding the predicted disease, is uploaded onto the Google firebase database where analysis on the data is carried out for deriving important information. It is a stand-alone model that works as personal health assistant and this model can be used for other disease management. As a future study, we intend to work on the data analysis (cloud) part of the model.

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